STA 431 INTRODUCTION TO PROBABILITY

Midterm II

Study Guide

Multivariate distributions

Mathematical expectation

Built around your exam topics

Freund Chapter 3, Sections 5–7, and Chapter 4, Sections 2–3 and 5–8. Concept explanations, worked examples, subsection checkpoints, and **70 original homework-style questions with worked solutions**.

Homework alignment. Includes complete worked keys for Homework 4 (common and STA 431 problems) and Homework 5 (common and STA 431 problems), with separately marked STA 631 extensions.

Companion: a two-sided, US Letter landscape notesheet designed to copy by hand, and the STAM2 calculator suite with guided solvers and an on-calculator encyclopedia.

Prepared for Ibrahim Shahid October 27, 2026

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Scope, reading plan, and notation

This guide follows your topic list. The repeated expectation item is combined into one topic; the seven resulting sections have ten original questions each. These questions model methods from Homeworks 4–5, rather than predicting the actual exam. The earlier whole-course study guide serves as a reading map; the textbook and homework PDFs supply the mathematical sources checked for this edition.

Study sequence. Read a concept and its example, cover the checkpoint answer, and solve the checkpoint. Then attempt the corresponding ten-question practice set without consulting the solutions. Use the homework keys to check whether you can reproduce the assigned methods. Finally work a mixed set: A2, A6, B4, C6, D2, D6, E4, F3, F8, G3.

Set	Skill	Homework connection
A	Joint, marginal, conditional distributions; support; probabilities	HW4 common 1–3; STA431 4–5
B	Expectation, functions of variables, linearity	HW5 common 1–2
C	Raw and central moments; variance	HW5 common 1–2
D	MGFs, derivatives, transformations, uniqueness	HW5 common 3–4
E	Product moments, covariance, correlation	HW5 STA431 8
F	One and two linear combinations	HW5 common 5–6; Freund Ch4 §7
G	Conditional mean and variance	HW5 common 7; STA431 8

Notation that prevents common errors

$p_{X,Y}$ denotes a joint PMF, $f_{X,Y}$ a joint PDF, and $F_{X,Y}$ a joint CDF. Freund often uses f for either a PMF or a PDF and g, h for marginals; the meaning depends on the support and whether you sum or integrate. A support is the set on which probability mass or density can be positive. Every density is zero outside its stated support unless otherwise specified.

$\mu_X = \mathbb{E}X$, $\sigma_X^2 = \mathrm{Var}(X)$, $\mu_r' = \mathbb{E}(X^r)$, and $\mu_r = \mathbb{E}[(X - \mu_X)^r]$. The quantity $\mathbb{E}(XY)$ is not generally $\mathbb{E}X\mathbb{E}Y$. In conditional formulas the value after the vertical bar is held fixed.

A repeatable solution method

1. Model: define the variables and translate words into an event or payoff. **2. Support:** list the pairs or sketch the region. **3. Normalize:** check nonnegativity and total mass one. **4. Choose:** marginal, conditional, probability, moment, or MGF. **5. Set up:** write the full sum/integral with limits before calculating. **6. Check:** probability in $[0, 1]$, variance nonnegative, sensible units, conditionals normalized.

Handwritten sheet. Copy the separate landscape master using its three-column layout on each side. The back prioritizes calculator workflows, with native commands and the STAM2 program entry point. The explanations and solutions stay here for learning. US Letter landscape is 11×8.5 inches; a 0.3 mm pen helps with small lettering, but handwriting size determines fit. The master uses approximately 8-point body text; use a printout at actual size as your spacing guide.

1 Multivariate random variables

Primary reading. Freund, Chapter 3, §5 "Multivariate Distributions," pp. 82–91; §6 "Marginal Distributions," pp. 92–95; §7 "Conditional Distributions," pp. 95–101. In Freund, $g(x)$ and $h(y)$ often denote the two marginals; here we write p_X, p_Y or f_X, f_Y to show which variable is retained. The word *distribution* can mean the entire probability law. A *distribution function* means a CDF; it is not the same object as a PMF or PDF.

> **Read with a purpose.** In §5, study Example 13 (p. 83) for normalization, Example 16 (pp. 86–87) for a full piecewise CDF, and the rectangle-increment task in Exercise 58 (p. 91). In §6, study Example 21 (p. 93) for marginal densities. In §7, compare Example 24 (pp. 96–97), where conditioning changes the law, with Example 25 (pp. 97–98), where it does not. This comparison explains independence more clearly than memorizing the product rule alone.

1.1 Joint laws, support, and normalization

A random vector (X, Y) records two numerical measurements from the same experiment. Its **support** describes the pairs that can occur. Write the support before calculating: the formula alone does not specify the model.

For discrete variables, the joint probability mass function is

$$p_{X,Y}(x, y) = P(X = x, Y = y), \qquad p_{X,Y}(x, y) \geq 0, \qquad \sum_x \sum_y p_{X,Y}(x, y) = 1.$$

The comma inside $P(X = x, Y = y)$ means "and." Sum only over allowed pairs. A probability table is a joint PMF written as a grid.

For an absolutely continuous random vector, the joint probability density satisfies

$$f_{X,Y}(x, y) \geq 0, \qquad \iint_{\mathbb{R}^2} f_{X,Y}(x, y)\, \mathrm{d}x\, \mathrm{d}y = 1, \qquad P((X, Y) \in A) = \iint_A f_{X,Y}(x, y)\, \mathrm{d}x\, \mathrm{d}y.$$

The density is zero off its support. A density is *probability per unit area*; its value need not be at most one. In an absolutely continuous model, $P(X = x, Y = y) = 0$, and lines such as $X = Y$ also have probability zero. The latter statement need not hold for a random vector concentrated on a line, even if each individual variable is continuous.

Finding an unknown constant. If $p(x, y) = c\, w(x, y)$ on a finite support S, then

$$c = \left[\sum_{(x,y) \in S} w(x, y) \right]^{-1}.$$

For a continuous density, replace the sum by the integral over S. Check nonnegativity as well as total mass. If the proposed formula changes sign, a normalizing constant alone cannot repair it.

Worked example: turn a formula into a joint table

Let $p(x,y) = c(1+x+2y)$ for $x \in \{0,1\}$ and $y \in \{0,1,2\}$, and zero otherwise. The six weights add to $1+3+5+2+4+6 = 21$, so $c = 1/21$:

	$y=0$	$y=1$	$y=2$	$p_X(x)$
$x=0$	$1/21$	$3/21$	$5/21$	$9/21$
$x=1$	$2/21$	$4/21$	$6/21$	$12/21$
$p_Y(y)$	$3/21$	$7/21$	$11/21$	1

For example, $P(X > Y) = p(1,0) = 2/21$. The event $X+Y \le 1$ contains exactly $(0,0),(0,1),(1,0)$, giving $P(X+Y \le 1) = 6/21 = 2/7$.

Checkpoint. Find c for $p(x,y) = cxy$ on $x \in \{1,2\}$, $y \in \{1,2\}$, and then find $P(X = Y)$. *Answer:* $c = 1/9$ and $P(X=Y) = (1+4)/9 = 5/9$.

1.2 Joint CDFs and recovering probabilities

For either type of random vector,

$$F_{X,Y}(x,y) = P(X \le x, Y \le y).$$

Thus a CDF gives the probability in the lower-left rectangle ending at (x,y). A CDF is defined for every real (x,y), takes values in $[0,1]$, is nondecreasing in either coordinate with the other fixed, and has nonnegative probability increments over rectangles. The last condition matters: being increasing in each coordinate alone is not enough to guarantee a valid joint CDF.

$$\text{Discrete:} \quad F_{X,Y}(x,y) = \sum_{s \le x} \sum_{t \le y} p_{X,Y}(s,t),$$

$$\text{Continuous:} \quad F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(s,t)\, \mathrm{d}t\, \mathrm{d}s.$$

For an absolutely continuous law, recover the density by

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x\, \partial y} \quad \text{almost everywhere.}$$

Differentiate only the appropriate branch and preserve its support. A density formula inside the support is not the full CDF.

Rectangle probabilities from a CDF. Inclusion–exclusion gives the identity

$$\boxed{P(a < X \le b,\ c < Y \le d) = F(b,d) - F(a,d) - F(b,c) + F(a,c).}$$

This identity holds for arbitrary laws. For a continuous joint density, changing strict to non-strict inequalities does not change the answer. For discrete variables, endpoints matter: if X, Y are integer valued,

$$P(a \le X \le b, c \le Y \le d) = F(b,d) - F(a-1,d) - F(b,c-1) + F(a-1,c-1)$$

for integer endpoints. At an isolated mass point, the PMF is the CDF's two-dimensional jump. On an integer grid,

$$p(x,y) = F(x,y) - F(x-1,y) - F(x,y-1) + F(x-1,y-1).$$

For a non-integer discrete support, use left limits at the point, rather than blindly subtracting one.

Worked example: a CDF with every boundary covered

Let

$$f(x,y) = \frac{3}{2}(x^2+y^2), \quad 0<x<1,\ 0<y<1, \qquad f(x,y)=0 \text{ otherwise.}$$

The integral of x^2+y^2 over the unit square is $2/3$, so this is normalized. In the interior,

$$F(x,y) = \int_0^x \int_0^y \frac{3}{2}(s^2+t^2)\ \mathrm{d}t\ \mathrm{d}s = \frac{1}{2}(x^3y+xy^3).$$

The integration must stop at the edge of the support. The complete answer is

$$F(x,y) = \begin{cases} 0, & x\le 0 \text{ or } y\le 0,\\ \frac{1}{2}(x^3y+xy^3), & 0<x<1,\ 0<y<1,\\ \frac{1}{2}(y+y^3), & x\ge 1,\ 0<y<1,\\ \frac{1}{2}(x+x^3), & 0<x<1,\ y\ge 1,\\ 1, & x\ge 1,\ y\ge 1. \end{cases}$$

All boundaries agree with their adjacent branches. A compact check is to set $u=\min(1,\max(0,x))$ and $v=\min(1,\max(0,y))$; then $F(x,y)=(u^3v+uv^3)/2$. In the interior, differentiating twice returns $\frac{3}{2}(x^2+y^2)$.

Checkpoint. For the CDF above, find $P(X\le 1/2, Y>1/2)$. *Answer:* $F(1/2,1)-F(1/2,1/2) = 5/16-1/16=1/4$.

1.3 Marginals: integrate or sum out the other variable

The marginal law of X describes X with Y unobserved. It is not obtained by inserting a convenient value of y into the joint law:

$$\text{Discrete:} \quad p_X(x)=\sum_y p_{X,Y}(x,y), \quad p_Y(y)=\sum_x p_{X,Y}(x,y),$$

$$\text{Continuous:} \quad f_X(x)=\int_{-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}y, \quad f_Y(y)=\int_{-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}x.$$

Keep the variable named in the subscript; eliminate the other one. The surviving variable is a parameter during the integration. State the marginal's support and put it equal to zero elsewhere.

From the joint CDF,

$$F_X(x)=\lim_{y\to\infty} F_{X,Y}(x,y), \qquad F_Y(y)=\lim_{x\to\infty} F_{X,Y}(x,y).$$

Once a marginal is known, compute its univariate probabilities by summing, integrating, or subtracting CDF values.

Worked example: support-dependent marginal bounds

Suppose

$$f(x,y)=8xy \quad \text{on } 0<x<y<1, \qquad f(x,y)=0 \text{ elsewhere.}$$

For fixed x, the allowed values of y run from x to 1. For fixed y, the allowed values of x run from 0 to y. Consequently,

$$f_X(x) = \int_x^1 8xy \, \mathrm{d}y = 4x(1-x^2), \quad 0 < x < 1,$$
$$f_Y(y) = \int_0^y 8xy \, \mathrm{d}x = 4y^3, \qquad 0 < y < 1.$$

Both densities integrate to one. The calculation also verifies that the original joint density is normalized. Using $\int_0^1$ for the first marginal would incorrectly include points with $y < x$ where the joint formula does not apply.

Checkpoint. For $f(x, y) = 2$ on $0 < x < y < 1$, find both marginals. *Answer:* $f_X(x) = 2(1-x)$ and $f_Y(y) = 2y$ on $(0, 1)$, zero otherwise.

1.4 Conditional laws and reconstructing a joint law

A conditional law describes the remaining uncertainty after one variable is fixed. Divide by the marginal of the *given* variable:

$$\boxed{p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}} \quad (p_Y(y) > 0), \qquad \boxed{f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}} \quad (f_Y(y) > 0).$$

For fixed y, sum or integrate over x and check that the answer is one. The conditional support is the slice of the original support at that y.

In a continuous model, $P(Y = y) = 0$. The conditional density formula defines the usual density-based conditional law; it is *not* literal division by the probability $P(Y = y)$. If the question gives an interval event such as $Y \in B$ instead, use

$$P(X \in A \mid Y \in B) = \frac{\int_B \int_A f_{X,Y}(x,y) \, \mathrm{d}x \, \mathrm{d}y}{\int_B f_Y(y) \, \mathrm{d}y}, \qquad P(Y \in B) > 0.$$

Restrict both integrals to the support. Conditioning on $Y \le 1/2$ is different from conditioning on $Y = 1/2$.

Conditional CDF. Once the conditional PMF or PDF is available,

$$F_{X|Y}(a \mid y) = \sum_{x \le a} p_{X|Y}(x \mid y) \quad \text{or} \quad F_{X|Y}(a \mid y) = \int_{-\infty}^a f_{X|Y}(x \mid y) \, \mathrm{d}x.$$

For a sufficiently smooth absolutely continuous law with $f_Y(y) > 0$,

$$F_{X|Y}(a \mid y) = \frac{\partial F_{X,Y}(a,y)/\partial y}{f_Y(y)}.$$

Integrating the conditional PDF is usually the safest route on an exam.

Worked example: a conditional slice

For the square density $\frac{3}{2}(x^2 + y^2)$ used above,

$$f_Y(y) = \frac{1}{2} + \frac{3}{2}y^2, \qquad f_{X|Y}(x \mid y) = \frac{3(x^2+y^2)}{1+3y^2}, \quad 0 < x < 1,$$

for $0 < y < 1$. Thus

$$P\left(X \le \frac{1}{2} \,\middle|\, Y = \frac{1}{2}\right) = \int_0^{1/2} \frac{12x^2 + 3}{7}\, \mathrm{d}x = \frac{2}{7}.$$

For the triangular density $8xy$, instead,

$$f_{Y|X}(y \mid x) = \frac{8xy}{4x(1-x^2)} = \frac{2y}{1-x^2}, \qquad x < y < 1, \quad 0 < x < 1.$$

Its support depends on x; ignoring that dependence would change the conditional law.

Which directions of conversion are possible?

- Joint law $\longrightarrow$ either marginal or either conditional: sum/integrate and divide, where the relevant denominator is positive.
- Marginal + matching conditional $\longrightarrow$ joint law:

  $$f_{X,Y}(x,y) = f_{X|Y}(x \mid y) f_Y(y) = f_{Y|X}(y \mid x) f_X(x),$$

  with the analogous PMF identities. Preserve both the marginal support and conditional slice support.
- Two marginals alone do *not* determine a joint law. They describe individual behavior but leave dependence unspecified. If independence is additionally established, multiply the marginals.
- One conditional law alone generally does not determine the joint law; the distribution of the conditioning variable is still missing.

For example, a Bernoulli(1/2) pair can be independent, equal almost surely ($Y = X$), or opposite almost surely ($Y = 1 - X$). Every case has the same two marginals, but the joint probabilities differ.

Checkpoint. Suppose $f_Y(y) = 2y$ on $(0,1)$ and, given $Y = y$, X is uniform on $(0, y)$. Find the joint density and f_X. *Answer:* $f_{X,Y}(x,y) = 2$ on $0 < x < y < 1$, and $f_X(x) = 2(1-x)$ on $(0,1)$.

1.5 Independence: what must be proved

Independence means that knowing either variable gives no information about the other:

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B) \quad \text{for every pair of measurable sets } A, B.$$

The following equivalent tests are useful in this course:

$$\begin{aligned} F_{X,Y}(x,y) &= F_X(x)F_Y(y) && \text{for every } x, y, \\ p_{X,Y}(x,y) &= p_X(x)p_Y(y) && \text{for every discrete pair,} \\ f_{X,Y}(x,y) &= f_X(x)f_Y(y) && \text{almost everywhere, for densities.} \end{aligned}$$

Equivalently, the conditional law equals the marginal law for all conditioning values of positive probability in the discrete case, or almost every conditioning value in the continuous case. One matching conditional slice does not prove independence.

Factorization and support. On a rectangular product support, a nonnegative normalized density that factors into a function of x times a function of y is independent. For example, $f(x,y) = 6xy^2$ on the unit square gives $f_X(x) = 2x$, $f_Y(y) = 3y^2$, and their product equals f. A rectangular support by itself does not establish independence: the square density $\frac{3}{2}(x^2 + y^2)$ is dependent.

For a density, equality or inequality at one isolated point has no probabilistic force, because a density can be changed there without altering the law. To show dependence, show a failure on a region of positive area, a genuine change in conditional distributions, or an event pair that violates the probability product rule.

Worked example: dependence caused by the support

For $f(x,y) = 8xy$ on $0 < x < y < 1$, the two marginal densities are positive throughout $(0,1)$. If X and Y were independent, the event

$$A = \{X > 3/4\}, \qquad B = \{Y < 1/4\}$$

would have positive probability $P(A)P(B)$. Yet $P(A \cap B) = 0$, since $x < y$ on the support. Thus X and Y are dependent. This argument uses whole intervals rather than individual density values.

Consequences worth knowing. Independent variables satisfy $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$ when the relevant expectations exist; functions of separately independent variables remain independent. Consequently, finite-variance independent variables have zero covariance, and variances of their sums add. Zero covariance alone does not imply independence. For three or more variables, pairwise independence alone does not imply mutual independence; multiplying all marginal laws requires mutual independence.

Checkpoint. Is $f(x,y) = 2$ on $0 < x < y < 1$ independent? Is $f(x,y) = 4xy$ on $0 < x < 1, 0 < y < 1$ independent? *Answer:* The triangular law is dependent; the square law is independent, with marginals $2x$ and $2y$.

1.6 Region probabilities and integration limits

The event and the support impose restrictions simultaneously. The desired region is always

$$R = \{\text{points satisfying the event}\} \cap \{\text{support of the joint law}\}.$$

Use the following order of work:

1. Sketch the support, including its boundary lines or curves.

2. Add the event boundary; shade the intersection.

3. Choose an integration order. The outer variable runs over the projection of the shaded region. For a fixed outer value, the inner variable runs from the entering boundary to the leaving boundary.

4. Split the integral wherever the entering or leaving boundary changes. Alternatively, use the complement if it gives a simpler region.

The differential nearest the integrand is integrated first. In $\int_a^b \int_{L(x)}^{U(x)} f(x,y)\, \mathrm{d}y\, \mathrm{d}x$, x is held fixed during the inner integration. Inner limits may depend on the outer variable; outer limits cannot depend on the inner variable.

Worked example: an event clipped by a triangle

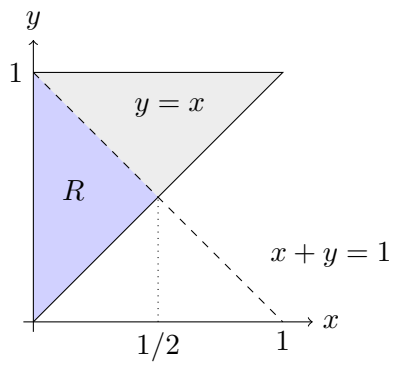
Use $f(x,y) = 8xy$ on $0 < x < y < 1$ and find $P(X+Y<1)$. The support requires $x < y$, while the event requires $y < 1-x$. Such a slice exists only when $x < 1-x$, or $x < 1/2$:

$$\begin{aligned} P(X+Y<1) &= \int_0^{1/2} \int_x^{1-x} 8xy \, \mathrm{d}y \, \mathrm{d}x \\ &= \int_0^{1/2} 4x\left[(1-x)^2 - x^2\right] \, \mathrm{d}x \\ &= \left[2x^2 - \frac{8}{3}x^3\right]_0^{1/2} = \frac{1}{6}. \end{aligned}$$

Reversing the order forces a split at $y = 1/2$:

$$P(X+Y<1) = \int_0^{1/2} \int_0^{y} 8xy \, \mathrm{d}x \, \mathrm{d}y + \int_{1/2}^{1} \int_0^{1-y} 8xy \, \mathrm{d}x \, \mathrm{d}y.$$

The two expressions describe exactly the same shaded set.



A rectangle can still require a split. On $0 < x < 1$, $0 < y < 2$, the event $X < Y$ can be written with one integral,

$$\int_0^1 \int_x^2 f(x,y) \, \mathrm{d}y \, \mathrm{d}x,$$

or with two integrals in the reverse order,

$$\int_0^1 \int_0^y f(x,y) \, \mathrm{d}x \, \mathrm{d}y + \int_1^2 \int_0^1 f(x,y) \, \mathrm{d}x \, \mathrm{d}y.$$

For $0 < y < 1$, the line $x = y$ is the right boundary; for $1 < y < 2$, the rectangle's boundary $x = 1$ takes over.

Checkpoint. Let $f(x,y) = 1/2$ on $0 < x < 1$, $0 < y < 2$. Write both orders and calculate $P(X+Y<1)$. *Answer:* $\int_0^1 \int_0^{1-x} \frac{1}{2} \, \mathrm{d}y \, \mathrm{d}x = \int_0^1 \int_0^{1-y} \frac{1}{2} \, \mathrm{d}x \, \mathrm{d}y = 1/4$.

1.7 Translating word problems into events

Name the random quantities and identify the logical operation before calculating. "And" is intersection; "either" usually includes the possibility that both occur unless explicitly excluded.

Words	Mathematical event or quantity
First component fails before the second	$X < Y$
Total cost is at most b	$X + Y \le b$
Costs differ by less than d	$\lvert X - Y \rvert < d$
At least one fails by time t	$\{X \le t\} \cup \{Y \le t\}$; equivalently $\min(X, Y) \le t$
Both fail by time t	$X \le t, Y \le t$; equivalently $\max(X, Y) \le t$
Both still operate at time t	$X > t, Y > t$
Reimbursement at least a when spending is x_0	$P(Y \ge a \mid X = x_0)$
Mean total payout after a policy change	$\mathbb{E}[aX + bY + c]$
Variability of that total	$\mathrm{Var}(aX + bY + c)$

Series and parallel lifetimes. A series device fails when *either* component fails, so its lifetime is $T = \min(X, Y)$. A parallel device continues operating until *both* have failed, so $T = \max(X, Y)$. Therefore,

$$\begin{aligned} P(\text{series failure by } t) &= 1 - P(X > t, Y > t) \\ &= F_X(t) + F_Y(t) - F_{X,Y}(t, t), \\ P(\text{parallel failure by } t) &= F_{X,Y}(t, t). \end{aligned}$$

Only if independence is established may the first expression be replaced by $1 - [1 - F_X(t)][1 - F_Y(t)]$, or the second by $F_X(t)F_Y(t)$.

Worked example: conditional reimbursement

Suppose spending X has density $f_X(x) = x/2$ for $0 < x < 2$ (hundreds of dollars), and reimbursement Y, given $X = x$, is uniform on $(0, x)$. Then

$$f_{Y|X}(y \mid x) = \frac{1}{x} \quad (0 < y < x), \qquad f_{X,Y}(x, y) = \frac{1}{2} \quad (0 < y < x < 2).$$

"Reimbursed at least \$100 after spending \$150" means

$$P(Y \ge 1 \mid X = 1.5) = \int_1^{1.5} \frac{1}{1.5} \, \mathrm{d}y = \frac{1}{3}.$$

The unit conversion must occur before selecting limits. The answer is conditional, so integrating the original joint density over a two-dimensional region would answer a different question.

Checkpoint. A device fails if either component fails. Express "the device lasts beyond two hours" and "the device fails within two hours" in terms of X, Y. *Answer:* The events are $\{X > 2, Y > 2\}$ and its complement $\{X \le 2\} \cup \{Y \le 2\}$.

Exam habits. Write support and event; normalize; derive only the required marginal; divide by the given-variable marginal if conditioning; state the formula before inserting numbers; and check that a probability lies in $[0, 1]$. On an independence question, write an actual equality proof or a specific probability-law counterexample.

Additional textbook practice, checked against Freund

The following are references to the supplied eighth edition; the task descriptions are paraphrases. Use the problem statements in the textbook for their exact numerical data.

- **Chapter 3, §5, Exercises 42–43, p. 90:** read events and CDF values from a joint probability table. This is direct preparation for Homework 4, Problem 4.
- **Chapter 3, §5, Exercises 44–45, p. 90:** first normalize a polynomial joint PMF, then add selected cells for events involving both variables.
- **Chapter 3, §5, Exercise 50, p. 90:** integrate a polynomial density over a smaller triangle inside triangular support; useful for practicing changing limits.
- **Chapter 3, §5, Exercises 54–55 and 58, p. 91:** recover a density from a CDF and calculate a rectangle probability; derive the four-term rectangle identity.
- **Chapter 3, §7, Exercises 69–70, p. 100:** obtain marginals and conditionals from a joint table. Exercise 70 uses the table in Exercise 42.
- **Chapter 3, §7, Exercises 73–75, p. 100:** justify independence for discrete laws, then derive marginals and conditional densities on a rectangle. Exercises 74–75 use the same density as Homework 4, Problem 2.
- **Chapter 3, §7, Exercises 76–77, p. 100:** derive marginals on nonrectangular supports and decide independence. Exercise 76 matches the form of the optional STA 631 triangular homework problem after interchanging x and y.
- **Chapter 3, §7, Exercise 79, p. 101:** recover a marginal CDF by letting the other CDF argument approach infinity.

2 Expectation: definitions, functions, and properties

Read: Freund Ch4 §2, pp.113–121; Bertsekas–Tsitsiklis Ch2 §2.4 and Ch3 §3.1; Wasserman §§3.1–3.2. Practice set B.

2.1 The definition and its meaning

Expectation is a probability-weighted average. A possible value receives weight equal to its probability in the discrete case; continuous values are weighted by a density:

$$\mathbb{E}X = \sum_x x p_X(x), \qquad \mathbb{E}X = \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x.$$

Sum or integrate over the support. A mean need not be a possible observation: a fair die has mean 3.5. It describes an average over repetitions, not the outcome you must see next.

A finite expectation is well defined when $\mathbb{E}|X| < \infty$. A nonnegative variable can have expectation $+\infty$; if the positive and negative contributions are both infinite, the expectation is undefined. Symmetry alone does not repair divergence.

Normalize before averaging

Let $p_X(x) = cx$ for $x = 1, 2, 3$. First $c(1+2+3) = 1$, so $c = 1/6$. Then $\mathbb{E}X = (1^2 + 2^2 + 3^2)/6 = 7/3$. The weights sum to one and the result lies between 1 and 3.

Practice checkpoint. If $f_X(x) = 2x$ on $(0, 1)$, find $\mathbb{E}X$.

Check: Integrate $\int_0^1 x(2x)\,\mathrm{d}x = 2/3$.

2.2 The generalized definition: LOTUS

For a function g, average the transformed values using the original probabilities:

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x), \qquad \mathbb{E}[g(X)] = \int g(x) f_X(x)\,\mathrm{d}x.$$

This is often called LOTUS, the law of the unconscious statistician. You do not need the distribution of $g(X)$ just to find its expectation. For two variables use the joint law:

$$\mathbb{E}[g(X,Y)] = \sum_x \sum_y g(x,y) p_{X,Y}(x,y) \quad \text{or} \quad \iint_S g(x,y) f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y.$$

For $\mathbb{E}[g(X)]$ alone a marginal suffices. For a product or nonlinear function of both variables, the marginals usually do not supply the needed dependence information.

A transformed payoff

With $f_X(x) = 2x$ on $(0, 1)$ and $Y = X^2$,

$$\mathbb{E}Y = \int_0^1 x^2(2x)\,\mathrm{d}x = \frac{1}{2}, \qquad \mathbb{E}(Y^2) = \mathbb{E}(X^4) = \int_0^1 2x^5\,\mathrm{d}x = \frac{1}{3}.$$

For a monetary payoff $R = 20X^2 - 3$, the expected net payoff is $20(1/2) - 3 = 7$ dollars.

Practice checkpoint. For $p(x) = x/6$, $x = 1, 2, 3$, find $\mathbb{E}[(X-1)^2]$.

Check: The sum is $(0 + 2 + 12)/6 = 7/3$, not $(\mathbb{E}X - 1)^2 = 16/9$.

2.3 Properties and how to use them

For integrable variables and constants,

$$\mathbb{E}c = c, \qquad \mathbb{E}(aX + bY + c) = a\mathbb{E}X + b\mathbb{E}Y + c,$$
$$X \le Y \text{ a.s.} \implies \mathbb{E}X \le \mathbb{E}Y, \qquad \mathbb{E}(\mathbf{1}_A) = P(A).$$

Linearity requires no independence: distribute the sum or integral across addition. If X, Y are independent and the expectations exist, then $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$. This product rule needs an assumption that linearity does not.

Nonlinearity

Generally $\mathbb{E}[g(X)] \neq g(\mathbb{E}X)$. In particular $\mathbb{E}(X^2) \neq (\mathbb{E}X)^2$ unless X is constant almost surely, and $\mathbb{E}(1/X) \neq 1/\mathbb{E}X$ in general. Expand before averaging: $\mathbb{E}[(aX + b)^2] = a^2\mathbb{E}X^2 + 2ab\mathbb{E}X + b^2$.

Practice checkpoint. If $\mathbb{E}X = 3$ and $\mathbb{E}Y = 5$ with arbitrary dependence, find $\mathbb{E}(2X - 4Y + 7)$. Can you obtain $\mathbb{E}(XY)$?

Check: The linear mean is $6 - 20 + 7 = -7$. The product mean is undetermined without more information.

3 Moments about the origin and the mean

Read: Freund Ch4 §3, pp.121–125; Wasserman §§3.1,3.3; Feller ChIX §§2–4. Practice set C.

3.1 Raw moments

The rth moment about the origin is $\mu'_r = \mathbb{E}(X^r)$, whenever finite. Thus $\mu'_0 = 1$, $\mu'_1 = \mu$, and $\mu'_2 = \mathbb{E}X^2$. Use LOTUS:

$$\mu'_r = \sum_x x^r p_X(x) \quad \text{or} \quad \int x^r f_X(x)\,\mathrm{d}x.$$

Odd powers can cancel across a symmetric distribution; even powers are nonnegative. For a symmetric density about zero with finite relevant moments, odd moments vanish. A finite fourth absolute moment implies finite moments of orders one through three.

A general moment formula

For $f_X(x) = 3x^2$ on $(0,1)$,

$$\mathbb{E}(X^r) = \int_0^1 3x^{r+2}\,\mathrm{d}x = \frac{3}{r+3} \quad (r > -3).$$

Hence $\mathbb{E}X = 3/4$, $\mathbb{E}X^2 = 3/5$, and $\mathbb{E}X^4 = 3/7$. One general formula replaces several nearly identical integrals.

Practice checkpoint. For X uniform on $(-2,2)$, find μ'_1, μ'_2, μ'_4.

Check: $0, 4/3, 16/5$, by symmetry and $\frac{1}{4}\int_{-2}^{2} x^r\,\mathrm{d}x$.

3.2 Central moments and variance

The rth moment about the mean is $\mu_r = \mathbb{E}[(X-\mu)^r]$. Special cases are

$$\mu_1 = 0, \qquad \mu_2 = \sigma^2 = \mathrm{Var}(X), \qquad \sigma = \sqrt{\mathrm{Var}(X)}.$$

Expanding the square gives

$$\mathrm{Var}(X) = \mathbb{E}(X^2 - 2\mu X + \mu^2) = \mathbb{E}X^2 - \mu^2.$$

Variance has squared units; standard deviation has the units of X. Also $\mathrm{Var}(X) \geq 0$, with equality precisely when $X = \mu$ almost surely. Expanding higher powers gives

$$\mu_3 = \mu'_3 - 3\mu\mu'_2 + 2\mu^3,$$
$$\mu_4 = \mu'_4 - 4\mu\mu'_3 + 6\mu^2\mu'_2 - 3\mu^4.$$

These formulas are optional conveniences; the definition always works.

Variance of a square needs a fourth moment

For the preceding $3x^2$ density,

$$\operatorname{Var}(X) = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{80}.$$

For $Y = X^2$,

$$\operatorname{Var}(Y) = \mathbb{E}X^4 - (\mathbb{E}X^2)^2 = \frac{3}{7} - \left(\frac{3}{5}\right)^2 = \frac{12}{175}.$$

Squaring $\operatorname{Var}(X)$ does not give $\operatorname{Var}(X^2)$.

Practice checkpoint. If $\mathbb{E}X = 2$, $\mathbb{E}X^2 = 7$, $\mathbb{E}X^3 = 28$, find $\operatorname{Var} X$ and μ_3.

Check: 3 and $28 - 3(2)(7) + 2(8) = 2$.

3.3 Shifts, scales, and standardization

For $Z = aX + b$, $Z - \mathbb{E}Z = a(X - \mathbb{E}X)$, so

$$\mathbb{E}Z = a\mu + b, \qquad \operatorname{Var} Z = a^2\sigma^2, \qquad \operatorname{SD}(Z) = |a|\sigma.$$

When $\sigma > 0$, $Z = (X - \mu)/\sigma$ has mean zero and variance one; standardization does not generally make a distribution normal. The rth central moment of $aX + b$ is $a^r\mu_r$.

Practice checkpoint. If $\mathbb{E}X = 10$, $\operatorname{Var} X = 9$, $Z = 5 - 2X$, find its mean, variance, and standard deviation.

Check: -15, 36, 6.

4 Moment generating functions

Read: Freund Ch4 §5, pp.127–131; Bertsekas–Tsitsiklis §4.1; Wasserman §3.6. Practice set D.

4.1 Definition, existence, and convergence

The MGF is $M_X(t) = \mathbb{E}(e^{tX})$, computed as

$$M_X(t) = \sum_x e^{tx} p_X(x) \quad \text{or} \quad M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x)\,\mathrm{d}x.$$

The variable is t; x is summed or integrated out. Always $M_X(0) = 1$. The standard moment and uniqueness theorems apply when the MGF is finite on an open interval containing zero. Being finite only at zero does not meet this hypothesis. Some distributions have moments but no MGF in such a neighborhood.

An exponential MGF and its domain

For $f_X(x) = \lambda e^{-\lambda x}$, $x > 0$, $\lambda > 0$,

$$M_X(t) = \lambda \int_0^{\infty} e^{-(\lambda - t)x}\,\mathrm{d}x = \frac{\lambda}{\lambda - t}, \qquad t < \lambda.$$

The original integral converges precisely when $\lambda - t > 0$. State this condition as part of the derivation.

Practice checkpoint. For $P(X = 0) = 1/3$, $P(X = 2) = 2/3$, find the MGF and domain.

Check: $M(t) = 1/3 + (2/3)e^{2t}$, for all real t.

4.2 Moments from derivatives or coefficients

Differentiating e^{tX} introduces powers of X:

$$M_X^{(r)}(0) = \mathbb{E}X^r, \qquad \operatorname{Var} X = M_X''(0) - [M_X'(0)]^2.$$

Equivalently,

$$M_X(t) = 1 + \mu t + \frac{\mathbb{E}X^2}{2!}t^2 + \frac{\mathbb{E}X^3}{3!}t^3 + \cdots.$$

Multiply the coefficient of t^r by $r!$ to obtain the raw moment. For the exponential example $M'(t) = \lambda/(\lambda - t)^2$ and $M''(t) = 2\lambda/(\lambda - t)^3$. The mean is $1/\lambda$, second moment $2/\lambda^2$, and variance $1/\lambda^2$.

The log-MGF $K(t) = \log M(t)$ is a useful shortcut:

$$K'(0) = \mathbb{E}X, \qquad K''(0) = \operatorname{Var} X.$$

This follows from $K' = M'/M$, $K'' = M''/M - (M'/M)^2$, and $M(0) = 1$.

Practice checkpoint. If $M(t) = 1 + 3t + 7t^2 + 10t^3 + O(t^4)$, find the mean, variance, and third raw moment.

Check: 3, $2(7) - 9 = 5$, and $3!\,10 = 60$.

4.3 Affine transformations and independent sums

For constants a, b,

$$M_{aX+b}(t) = \mathbb{E}[e^{t(aX+b)}] = e^{bt} M_X(at).$$

For $Z = (X - c)/d$, $d \neq 0$, this becomes

$$M_Z(t) = e^{-ct/d} M_X(t/d).$$

Substitute into every occurrence of the old argument. The transformed domain consists of t for which at belongs to the original domain. A negative scale reverses inequalities when solving for t.

For mutually independent $X_1, \ldots, X_n$,

$$M_{\sum_i a_i X_i + b}(t) = e^{bt} \prod_i M_{X_i}(a_i t).$$

This product rule needs independence of the exponential factors.

Centering and scaling

If $M_X(t) = e^{6t+18t^2}$ and $Z = (X - 6)/6$, then

$$M_Z(t) = e^{-t} e^{6(t/6)+18(t/6)^2} = e^{t^2/2}.$$

Thus $K_Z(t) = t^2/2$, $\mathbb{E}Z = 0$, and $\operatorname{Var} Z = 1$.

Practice checkpoint. If $M_X(t) = 2/(2 - t)$ for $t < 2$ and $Z = -3X + 1$, give its MGF and domain.

Check: $M_Z(t) = 2e^t/(2 + 3t)$ for $t > -2/3$.

4.4 Uniqueness and recognition

If two MGFs are finite and agree on an open interval containing zero, the variables have the same distribution. Matching only the first two moments does not identify a law.

Law	MGF	Mean; variance
Bernoulli(p)	$1 - p + pe^t$	p; $p(1-p)$
Binomial(n, p)	$(1 - p + pe^t)^n$	np; $np(1-p)$
Exponential, rate λ	$\lambda/(\lambda - t)$, $t < \lambda$	$1/\lambda$; $1/\lambda^2$
Normal(μ, σ^2)	$e^{\mu t + \sigma^2 t^2/2}$	μ; σ^2

Recognition is secondary here; derivatives give the required moments directly. In the normal form the coefficient of t^2 is half the variance. Freund's exponential distribution elsewhere uses a scale; here λ is explicitly a rate.

Practice checkpoint. Two variables have MGF $(1/4 + 3e^t/4)^4$ near zero. What follows? What if only their means and variances match?

Check: The MGFs identify the same binomial$(4, 3/4)$ law. Equal means and variances alone do not identify a law.

5 Product moments, covariance, and correlation

Reading. Freund, Chapter 4, §6, pp. 131–135, defines product moments and covariance and proves the independence implication. For the general correlation coefficient, use Bertsekas–Tsitsiklis, §4.5, chapter-local pp. 29–31. Freund also discusses correlation in Chapter 6, §7, p. 196, in the bivariate-normal setting.

5.1 Raw and central product moments

Product moments describe two variables together. The (r, s) product moment about the origin is

$$\mu'_{r,s} = \mathbb{E}[X^r Y^s] = \sum_x \sum_y x^r y^s p(x, y) \quad \text{or} \quad \iint x^r y^s f(x, y)\, \mathrm{d}x\, \mathrm{d}y.$$

Here r, s are nonnegative integers and the sum or integral is over the joint support. The corresponding central product moment is

$$\mu_{r,s} = \mathbb{E}[(X - \mu_X)^r (Y - \mu_Y)^s].$$

Special cases connect this notation with familiar quantities:

$$\mu'_{1,0} = \mathbb{E}[X], \quad \mu'_{0,1} = \mathbb{E}[Y], \quad \mu_{2,0} = \mathrm{Var}(X), \quad \mu_{0,2} = \mathrm{Var}(Y), \quad \mu_{1,1} = \mathrm{Cov}(X, Y).$$

In particular, $\mathbb{E}[XY]$ is a *raw* product moment; it is not usually the covariance. Existence matters: finite second moments of both variables guarantee that $\mathbb{E}|XY|$ is finite by Cauchy–Schwarz, so covariance is well defined.

Checkpoint. What are $\mu'_{2,1}$ and $\mu_{1,0}$? *Answer:* $\mathbb{E}[X^2 Y]$ and $\mathbb{E}[X - \mu_X] = 0$, respectively, when the expectations exist.

5.2 Covariance: definition, calculation, and properties

Covariance measures signed linear association:

$$\boxed{\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].}$$

To derive the computational formula, expand the centered product and use linearity:

$$\begin{aligned} \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] &= \mathbb{E}[XY] - \mu_X \mathbb{E}[Y] - \mu_Y \mathbb{E}[X] + \mu_X \mu_Y \\ &= \mathbb{E}[XY] - \mu_X \mu_Y. \end{aligned}$$

For hand calculations, find $\mathbb{E}[X]$, $\mathbb{E}[Y]$, and $\mathbb{E}[XY]$ before subtracting. Compute $\mathbb{E}[XY]$ from the joint law, not by multiplying marginal expectations unless independence has been established.

Worked example: a small probability table

Suppose the joint probabilities are

	$Y = 0$	$Y = 1$	$p_X(x)$
$X = 0$	1/10	2/10	3/10
$X = 1$	3/10	4/10	7/10
$p_Y(y)$	4/10	6/10	1

Because both variables take only values zero and one,

$$\mathbb{E}[X] = \frac{7}{10}, \qquad \mathbb{E}[Y] = \frac{3}{5}, \qquad \mathbb{E}[XY] = 1 \cdot 1 \cdot \frac{4}{10} = \frac{2}{5}.$$

Thus $\operatorname{Cov}(X, Y) = 2/5 - (7/10)(3/5) = -1/50$. Also, $\operatorname{Var}(X) = 21/100$ and $\operatorname{Var}(Y) = 6/25$.

Algebra to use freely. For finite second moments,

$$\begin{aligned} \operatorname{Cov}(X, Y) &= \operatorname{Cov}(Y, X), & \operatorname{Cov}(X, X) &= \operatorname{Var}(X), \\ \operatorname{Cov}(X, c) &= 0, & \operatorname{Cov}(aX + b, cY + d) &= ac \operatorname{Cov}(X, Y), \\ \operatorname{Cov}(X, Y + Z) &= \operatorname{Cov}(X, Y) + \operatorname{Cov}(X, Z). \end{aligned}$$

Shifts disappear because they disappear after centering. Scaling multiplies covariance by one factor from each variable; covariance need not be nonnegative.

Checkpoint. If $\operatorname{Cov}(X, Y) = -2$, find $\operatorname{Cov}(3X + 7, -2Y + 5)$. *Answer:* $3(-2)(-2) = 12$.

5.3 Correlation: normalized linear association

When $0 < \operatorname{Var}(X), \operatorname{Var}(Y) < \infty$, define

$$\boxed{\rho_{X,Y} = \operatorname{Corr}(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}, \qquad -1 \le \rho_{X,Y} \le 1.}$$

Dividing by standard deviations removes the units. Cauchy–Schwarz gives $|\operatorname{Cov}(X, Y)| \le \sigma_X \sigma_Y$, which proves the bound. Correlation is undefined if either variance is zero; a constant variable has zero covariance with every finite-variance variable, but it does not have a defined correlation coefficient.

For the preceding table,

$$\rho = \frac{-1/50}{\sqrt{(21/100)(6/25)}} = -\frac{1}{3\sqrt{14}} \approx -0.0891.$$

The small value indicates weak linear association, not independence. If $Y = aX + b$ almost surely with $a \neq 0$ and $\operatorname{Var}(X) > 0$, then $\rho = \operatorname{sgn}(a)$. Conversely, $|\rho| = 1$ implies such an affine relation almost surely. For nonzero a, c,

$$\operatorname{Corr}(aX + b, cY + d) = \operatorname{sgn}(ac) \operatorname{Corr}(X, Y).$$

Changing units by positive scale factors leaves correlation unchanged; reversing one scale reverses its sign.

Checkpoint. If $Y = 8 - 3X$ and $\operatorname{Var}(X) > 0$, find $\operatorname{Cov}(X, Y)$ and $\operatorname{Corr}(X, Y)$. *Answer:* $-3 \operatorname{Var}(X)$ and -1.

5.4 Independence implies zero covariance; the converse fails

If X, Y are independent and have finite second moments, factorization yields

$$\mathbb{E}[XY] = \iint xy f_X(x) f_Y(y) \, \mathrm{d}x \, \mathrm{d}y = \mathbb{E}[X]\mathbb{E}[Y], \qquad \operatorname{Cov}(X, Y) = 0.$$

The discrete proof uses two sums. More generally, $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$ under suitable integrability. Therefore, nonzero covariance disproves independence. Zero covariance only says that this one centered product has mean zero; other forms of dependence may remain.

Worked example: zero covariance with exact dependence

Let X be uniform on $(-1, 1)$ and let $Y = X^2$. Symmetry gives

$$\mathbb{E}[X] = \mathbb{E}[X^3] = 0, \quad \mathbb{E}[Y] = \mathbb{E}[X^2] = \frac{1}{3}, \quad \mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0.$$

They are nevertheless dependent. Set $A = \{|X| < 1/2\}$ and $B = \{Y < 1/4\}$. Since $Y = X^2$, these are the same event, and

$$P(A \cap B) = \frac{1}{2} \neq \frac{1}{4} = P(A)P(B).$$

Both variables have positive variance (1/3 and 4/45), so their correlation is defined and equals zero. This pair lies on a curve and has no joint density with respect to planar area; compute its expectations from the given univariate law.

Checkpoint. Which conclusion is justified: (i) $\rho = 0$ implies independence; (ii) independence with positive finite variances implies $\rho = 0$? *Answer:* Only (ii). A jointly normal pair has the additional special property that zero correlation does imply independence, but that conclusion requires the joint-normal assumption.

Textbook practice. Freund Chapter 4, Exercise 42 (p. 140): covariance from a four-point joint law; Exercises 46–47 (p. 140): zero covariance with dependence. Exercise 47 uses the same form as the optional STA 631 part of Homework 5.

6 Moments of linear combinations

Reading. Freund, Chapter 4, §7, pp. 136–138; Theorem 14 treats one linear combination, and Theorem 15 treats the covariance of two linear combinations. The examples below use original numerical data. Assume finite second moments whenever a variance or covariance is used.

6.1 One linear combination: mean and variance

For $W = c + \sum_{i=1}^{n} a_i X_i$, linearity gives

$$\boxed{\mathbb{E}[W] = c + \sum_{i=1}^{n} a_i \mathbb{E}[X_i].}$$

This formula does not require independence. To calculate variance, center W:

$$W - \mathbb{E}[W] = \sum_{i} a_i (X_i - \mu_i).$$

Squaring creates one square term per variable and two identical cross terms per pair. Taking expectations therefore gives

$$\boxed{\operatorname{Var}(W) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(X_i) + 2 \sum_{i<j} a_i a_j \operatorname{Cov}(X_i, X_j).}$$

The constant c has vanished. The coefficients in the variance terms are squared, but the coefficients in the covariance terms retain their signs.

For two variables,

$$\operatorname{Var}(aX + bY + c) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}.$$

In particular,

$$\operatorname{Var}(X + Y) = \sigma_X^2 + \sigma_Y^2 + 2\sigma_{XY},$$
$$\operatorname{Var}(X - Y) = \sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}.$$

The variance of a difference still contains the *sum* of the two individual variances. Only the covariance term changes sign. If the variables are independent, or merely pairwise uncorrelated, all cross-covariance terms vanish. Do not remove these terms without such information.

Worked example: a changed total

Suppose $\mathbb{E}[X] = 1$, $\mathbb{E}[Y] = 2$, $\operatorname{Var}(X) = 4$, $\operatorname{Var}(Y) = 9$, and $\operatorname{Cov}(X, Y) = -1$. For $W = 3X - 2Y + 7$,

$$\mathbb{E}[W] = 3(1) - 2(2) + 7 = 6,$$

$$\operatorname{Var}(W) = 3^2(4) + (-2)^2(9) + 2(3)(-2)(-1) = 84.$$

The negative coefficient and negative covariance combine to make the cross term positive. The fixed payment 7 affects the mean but not the variance.

Checkpoint. If X, Y are independent with variances 2 and 5, find $\operatorname{Var}(4X - Y + 100)$. *Answer:* $16(2) + 5 = 37$.

6.2 Three or more variables: count each pair once

For $W = aX + bY + cZ + d$, write all six terms before substituting:

$$\begin{aligned}\operatorname{Var}(W) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + c^2 \operatorname{Var}(Z) \\ + 2ab \operatorname{Cov}(X,Y) + 2ac \operatorname{Cov}(X,Z) + 2bc \operatorname{Cov}(Y,Z).\end{aligned}$$

For n variables there are $n(n-1)/2$ distinct pairs. This is why the notation $\sum_{i<j}$ is useful: it counts each pair once and supplies the factor two outside. The equivalent full double sum $\sum_i \sum_j a_i a_j \operatorname{Cov}(X_i, X_j)$ already counts both orders; do not add another factor two to it.

Worked example: track the signs systematically

Suppose

$$\begin{gathered}\operatorname{Var}(X) = 4, \quad \operatorname{Var}(Y) = 9, \quad \operatorname{Var}(Z) = 1, \\ \operatorname{Cov}(X,Y) = 1, \quad \operatorname{Cov}(X,Z) = 0, \quad \operatorname{Cov}(Y,Z) = -1.\end{gathered}$$

For $W = 2X - Y + 3Z + 5$,

$$\begin{aligned}\operatorname{Var}(W) &= 4(4) + 1(9) + 9(1) \\ &\quad + 2(2)(-1)(1) + 2(2)(3)(0) + 2(-1)(3)(-1) \\ &= 16 + 9 + 9 - 4 + 0 + 6 = 36.\end{aligned}$$

If the means are $1, 2, -1$, respectively, then $\mathbb{E}[W] = 2(1) - 2 + 3(-1) + 5 = 2$.

Checkpoint. For independent identically distributed $X_1, \ldots, X_n$ with mean μ and variance σ^2, find the mean and variance of $\bar{X} = n^{-1} \sum_i X_i$. *Answer:* $\mathbb{E}[\bar{X}] = \mu$ and $\operatorname{Var}(\bar{X}) = \sigma^2/n$. The same variance formula holds if the variables have the common variance and are pairwise uncorrelated.

6.3 Two linear combinations: covariance

Let $U = c + \sum_i a_i X_i$ and $V = d + \sum_i b_i X_i$. Bilinearity of covariance gives the universal formula

$$\boxed{\operatorname{Cov}(U,V) = \sum_i \sum_j a_i b_j \operatorname{Cov}(X_i, X_j).}$$

Separating diagonal terms from off-diagonal terms gives Freund's equivalent form:

$$\operatorname{Cov}(U,V) = \sum_i a_i b_i \operatorname{Var}(X_i) + \sum_{i<j} (a_i b_j + a_j b_i) \operatorname{Cov}(X_i, X_j).$$

Here the paired coefficient is $a_i b_j + a_j b_i$, not generally $2a_i b_j$. Both fixed constants disappear. For two underlying variables,

$$\operatorname{Cov}(aX + bY, cX + dY) = ac \operatorname{Var}(X) + bd \operatorname{Var}(Y) + (ad + bc) \operatorname{Cov}(X,Y).$$

Worked example: two totals built from the same variables

Use $\operatorname{Var}(X) = 4$, $\operatorname{Var}(Y) = 9$, $\operatorname{Cov}(X,Y) = -1$, and let $U = 2X - Y + 1$, $V = X + 3Y - 4$. Then

$$\begin{aligned}\operatorname{Cov}(U,V) &= 2(1)(4) + (-1)(3)(9) + [2(3) + (-1)(1)](-1) \\ &= 8 - 27 - 5 = -24.\end{aligned}$$

Separately, $\mathrm{Var}(U) = 29$ and $\mathrm{Var}(V) = 79$, giving $\mathrm{Corr}(U, V) = -24/\sqrt{29 \cdot 79}$. A useful internal check is

$$\mathrm{Var}(U + V) = 29 + 79 + 2(-24) = 60.$$

Since $U + V = 3X + 2Y - 3$, direct calculation gives $9(4) + 4(9) + 12(-1) = 60$, agreeing exactly.

Even if the underlying X_i are independent, two combinations built from the same variables need not be independent or uncorrelated. Underlying independence reduces their covariance to $\sum_i a_i b_i \mathrm{Var}(X_i)$; it does not make this diagonal sum disappear.

Checkpoint. Show that $\mathrm{Cov}(X + Y, X - Y) = \mathrm{Var}(X) - \mathrm{Var}(Y)$, without assuming independence. *Answer:* Expand by bilinearity; $-\mathrm{Cov}(X, Y) + \mathrm{Cov}(Y, X) = 0$. Thus equal variances make the sum and difference uncorrelated, but do not establish their independence.

Textbook practice. Freund Chapter 4, Exercises 49–50 (p. 140) compare means and variances with and without independence. Exercises 53–55 (p. 140) practice the covariance of two combinations, including the sum-and-difference identity.

7 Conditional expectation and conditional variance

Reading. Freund, Chapter 4, §8, pp. 138–139, Definition 10 and Example 20. The final subsection on averaging conditional means and variances is supplementary reading from Bertsekas–Tsitsiklis, §4.3, chapter-local pp. 17–24; it strengthens the listed exam topics without replacing the direct calculations required by Homework 5.

7.1 Definition: take an expectation under the conditional law

Once $Y = y$ is given, use the distribution of $X \mid Y = y$ as the probability weights. For a function g,

$$\boxed{\mathbb{E}[g(X) \mid Y = y] = \sum_x g(x) p_{X|Y}(x \mid y)}$$

in the discrete case, or

$$\boxed{\mathbb{E}[g(X) \mid Y = y] = \int g(x) f_{X|Y}(x \mid y)\, \mathrm{d}x}$$

in the continuous case. The limits are the conditional support. Treat y as a fixed parameter while summing or integrating over x. The choices $g(x) = x$ and $g(x) = x^2$ give the conditional first and second raw moments.

Substituting the conditional density formula gives a convenient quotient:

$$\mathbb{E}[g(X) \mid Y = y] = \frac{\displaystyle\int g(x) f_{X,Y}(x, y)\, \mathrm{d}x}{f_Y(y)}, \qquad f_Y(y) > 0.$$

Use a sum and $p_Y(y) > 0$ in the discrete version. This formula can save writing a separate conditional density, but the denominator and support must still be correct. The numerator alone is generally not a conditional expectation.

Worked example: a conditional mean from a table

Suppose

	$Y = 0$	$Y = 1$	$p_X(x)$
$X = 1$	$1/8$	$3/8$	$1/2$
$X = 2$	$3/8$	$1/8$	$1/2$
$p_Y(y)$	$1/2$	$1/2$	1

Given $Y = 0$, divide the first column by $1/2$, obtaining conditional probabilities $1/4, 3/4$. Thus

$$\mathbb{E}[X \mid Y = 0] = 1\left(\frac{1}{4}\right) + 2\left(\frac{3}{4}\right) = \frac{7}{4}, \qquad \mathbb{E}[X^2 \mid Y = 0] = \frac{1}{4} + 4\left(\frac{3}{4}\right) = \frac{13}{4}.$$

Given $Y = 1$, the conditional probabilities reverse, so $\mathbb{E}[X \mid Y = 1] = 5/4$.

Checkpoint. If $P(X = 0 \mid Y = 2) = 1/3$ and $P(X = 3 \mid Y = 2) = 2/3$, find $\mathbb{E}[X \mid Y = 2]$ and $\mathbb{E}[X^2 \mid Y = 2]$. *Answer:* 2 and 6. Squaring the conditional mean would give 4, which is a different quantity.

7.2 Conditional variance and support-dependent calculations

The conditional mean is $m(y) = \mathbb{E}[X \mid Y = y]$. The conditional variance measures spread around that conditional mean:

$$\boxed{\operatorname{Var}(X \mid Y = y) = \mathbb{E}[X^2 \mid Y = y] - \{\mathbb{E}[X \mid Y = y]\}^2.}$$

This follows by expanding $\mathbb{E}[(X - m(y))^2 \mid Y = y]$, treating $m(y)$ as constant. It is not $\mathbb{E}[X^2 \mid Y = y] - \{\mathbb{E}[X]\}^2$: use the same conditional law for both terms.

For the preceding table,

$$\operatorname{Var}(X \mid Y = 0) = \frac{13}{4} - \left(\frac{7}{4}\right)^2 = \frac{3}{16}.$$

Given $Y = 1$, the variance is also $3/16$, although the mean differs. Equal conditional variances do not imply independence.

Worked example: a conditional density on a moving interval

Let $f(x, y) = 3y$ on $0 < x < y < 1$, zero elsewhere. Integrate over the slice in x:

$$f_Y(y) = \int_0^y 3y \, \mathrm{d}x = 3y^2, \qquad f_{X|Y}(x \mid y) = \frac{1}{y}, \quad 0 < x < y, \quad 0 < y < 1.$$

Thus $X \mid Y = y$ is uniform on $(0, y)$. Direct calculation gives

$$\mathbb{E}[X \mid Y = y] = \int_0^y \frac{x}{y} \, \mathrm{d}x = \frac{y}{2}, \qquad \mathbb{E}[X^2 \mid Y = y] = \int_0^y \frac{x^2}{y} \, \mathrm{d}x = \frac{y^2}{3},$$

and therefore

$$\operatorname{Var}(X \mid Y = y) = \frac{y^2}{3} - \frac{y^2}{4} = \frac{y^2}{12}.$$

At $y = 1/2$, the conditional mean is $1/4$ and the conditional variance is $1/48$. Replacing the upper limit y by 1 would fail to normalize the conditional law and produce incorrect moments.

Checkpoint. In this triangular model, find $\mathbb{E}[X^2 \mid Y = y]$ and $\operatorname{Var}(X^2 \mid Y = y)$. *Answer:* $y^2/3$ and $y^4/5 - (y^2/3)^2 = 4y^4/45$; the second calculation needs the conditional fourth moment of X.

7.3 A function of a value versus a random variable

The expression $\mathbb{E}[X \mid Y = y] = m(y)$ is a number once y is fixed. The expression $\mathbb{E}[X \mid Y] = m(Y)$ is a random variable, because its value depends on the unobserved outcome of Y.

In the triangular example, $m(y) = y/2$, so $\mathbb{E}[X \mid Y] = Y/2$, while $\mathbb{E}[X \mid Y = 1/2] = 1/4$. In the table example, $m(Y)$ takes values $7/4$ and $5/4$, each with probability $1/2$.

Conditional expectation remains linear:

$$\mathbb{E}[aX + b \mid Y = y] = a\mathbb{E}[X \mid Y = y] + b.$$

Likewise $\operatorname{Var}(aX + b \mid Y = y) = a^2 \operatorname{Var}(X \mid Y = y)$. A quantity determined by the given variable can be treated as fixed inside the conditional expectation; for example, $\mathbb{E}[Yg(X) \mid Y = y] = y\mathbb{E}[g(X) \mid Y = y]$. With independence and integrability, $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$ almost surely. A constant conditional mean alone does not prove independence, since higher conditional moments or other distributional features may still change.

Checkpoint. If $\mathbb{E}[X \mid Y = y] = 2y - 1$, find $\mathbb{E}[3X + 4 \mid Y = 2]$ and express $\mathbb{E}[X \mid Y]$. *Answer:* 13 and $2Y - 1$.

7.4 Supplement: averaging conditional means and variances

The law of total expectation, also called the tower rule, says

$$\boxed{\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].}$$

In a continuous model, this is $\mathbb{E}[X] = \int m(y) f_Y(y)\, \mathrm{d}y$. The reason is that the conditional expectation's numerator, multiplied by $f_Y(y)$ and integrated in y, reconstructs the joint-law integral for $\mathbb{E}[X]$.

For finite second moments, the law of total variance is

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]).}$$

The first term is the average spread within conditional groups; the second is the spread between their means. To derive it, average the conditional identity $\mathrm{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - m(Y)^2$ and add $\mathrm{Var}(m(Y)) = \mathbb{E}[m(Y)^2] - \{\mathbb{E}[X]\}^2$.

Worked check: recover unconditional moments

For $f(x, y) = 3y$ on $0 < x < y < 1$, $f_Y(y) = 3y^2$, so

$$\mathbb{E}[Y] = \frac{3}{4}, \qquad \mathbb{E}[Y^2] = \frac{3}{5}, \qquad \mathrm{Var}(Y) = \frac{3}{80}.$$

Using the conditional results,

$$\mathbb{E}[X] = \mathbb{E}[Y/2] = \frac{3}{8},$$

$$\mathrm{Var}(X) = \mathbb{E}[Y^2/12] + \mathrm{Var}(Y/2) = \frac{1}{20} + \frac{3}{320} = \frac{19}{320}.$$

The direct check is $\mathbb{E}[X^2] = \mathbb{E}[Y^2/3] = 1/5$, and $1/5 - (3/8)^2 = 19/320$.

Checkpoint. In the discrete table above, recover $\mathbb{E}[X]$ and $\mathrm{Var}(X)$ using the two total laws. *Answer:* $\mathbb{E}[X] = (7/4 + 5/4)/2 = 3/2$. The average conditional variance is $3/16$ and the variance of the two conditional means is $1/16$, so $\mathrm{Var}(X) = 1/4$.

Textbook practice. Freund Chapter 4, Example 20 (p. 139), gives the direct conditional-mean and conditional-variance workflow used in Homework 5, Problem 7. Exercise 60 (p. 140) conditions a univariate law on an interval; it is useful additional practice in recognizing the conditioning event and normalizing correctly.

8 Homework 4: every problem with STAM2

This walkthrough covers the three common questions, both STA 431 questions, and the two separately labeled STA 631 questions. Start with `PRGM > STAM2`. The routes below name the actual solver menus. For the first five questions, use `HOMEWORK MAP > LOAD + SOLVE`, then `HW4 COMMON +431` and the matching question. This preloads inputs, not answers: the general solvers still calculate each result. Manual entries below show the model being solved and let you handle altered exam numbers.

For continuous calculations, a displayed fraction is a numerical result that suggests an exact value. The integral and its algebra establish the exact answer. In particular, the app does not symbolically derive a marginal-density formula or prove independence of continuous variables.

8.1 Common Question 1: joint PMF, marginals, conditioning, independence

The joint PMF is $p(x, y) = c(x + y^2)$ on $x \in \{1, 2\}$ and $y \in \{1, 2, 3\}$.

Quick load: choose `Q1 JOINT PMF`; continue past its model screen.

Manual alternative: load the model once. Choose `DISCRETE + JOINT > FORMULA/GRID`:

Prompt	Enter
`x VALUES LIST?`	`{1,2}`
`y VALUES LIST?`	`{1,2,3}`
`q(X,Y)?`	`X+Y^2`

The app constructs all six pairs, sums their weights to 37, and displays $c = 1/37$. You do not have to enter c or manually type the six joint probabilities.

(a) What to write. The six weights are $2, 5, 10, 3, 6, 11$, hence $1 = c(37)$ and $\boxed{c = 1/37}$.

(b) Marginals. From `TABLE TOOLS`, choose `MARG/CONDITIONAL`. Choose the new `MARGINAL TABLES` option to display both complete marginal tables. Press `ENTER` between them. You no longer have to query the five values individually. The returned values are

x	1	2
$p_X(x)$	17/37	20/37

y	1	2	3
$p_Y(y)$	5/37	11/37	21/37.

For the general formulas, write

$$p_X(x) = \sum_{y=1}^{3} \frac{x + y^2}{37} = \frac{3x + 14}{37}, \quad x = 1, 2;$$

$$p_Y(y) = \sum_{x=1}^{2} \frac{x + y^2}{37} = \frac{3 + 2y^2}{37}, \quad y = 1, 2, 3.$$

Both PMFs are zero outside their supports.

(c) Conditional PMF. Choose `X GIVEN Y` and enter 2. The conditional-distribution screen gives

$$\boxed{p_{X|Y}(1 \mid 2) = 5/11, \qquad p_{X|Y}(2 \mid 2) = 6/11.}$$

The written step is $(5/37)/(11/37) = 5/11$ and $(6/37)/(11/37) = 6/11$. These sum to one. Extra conditional moments shown afterward are not required by this question.

(d) Independence. Choose `INDEPENDENCE`. The app reports a numerical factorization failure and now displays a violating pair with its joint probability and marginal product. Convert that observation into an exact comparison. For example:

$$p(1,1) = \frac{2}{37} = \frac{74}{1369} \neq \frac{17}{37}\frac{5}{37} = \frac{85}{1369} = p_X(1)p_Y(1).$$

Therefore X, Y are dependent. One failing PMF pair is sufficient.

Why this is easier. The homework loader removes model re-entry; the manual alternative needs just three model entries. The new marginal-table command displays all five marginal values together, and the independence tool supplies a counterexample pair. The same data stay loaded for all subparts. You still write the marginal sums and the exact independence comparison.

8.2 Common Question 2: a joint PDF and conditional-density formula

Here $f(x, y) = c(2x + y)$ on $0 < x < 1$, $0 < y < 2$.

Quick load: choose `Q2 JOINT PDF`, then enter outer resolution $N = 4$.

(a) Manual normalization route. Choose `DENSITY + INTEGRAL > DOUBLE INTEGRAL`. Enter the following model. The fixed parameter A is unused, so enter 0.

Prompt	Enter
`q(X,Y)?`	`2X+Y`
`FIXED A?`	`0`
`LOWER X?, UPPER X?`	`0, 1`
`LOWER Y(X)?, UPPER Y(X)?`	`0, 2`
`g(X,Y),1=AREA?`	`1`
`EVEN N 4-100?`	`4`

The app displays raw integral $I \approx 4$ at $N = 4$ and $N = 8$. Choose `NORMALIZE FULL S`; because the entered rectangle is the full support, the app gives $c \approx 1/4$.

What to write for (a).

$$1 = c \int_0^1 \int_0^2 (2x + y)\, dy\, dx = c \int_0^1 (4x + 2)\, dx = 4c, \qquad \boxed{c = \frac{1}{4}}.$$

The numerical solver removes the arithmetic burden, while this short calculation establishes the exact constant.

(b) Derive the marginal of Y. Hold y fixed and integrate over x:

$$\boxed{f_Y(y) = \int_0^1 \frac{2x + y}{4}\, dx = \frac{1 + y}{4}, \quad 0 < y < 2;}$$

$f_Y(y) = 0$ otherwise. The app can check values of this function, but it cannot return the symbolic expression for all y.

Numerical check of (b), using the improved joint-slice tool. Choose `DENSITY + INTEGRAL > JOINT SLICE > X GIVEN Y`. Enter joint shape `2X+Y`, known $c = 1/4$, given value 1, and remaining-variable bounds 0 and 1. The results screen reports marginal density $f_Y(1) \approx 1/2$. To check another y, use `NEW GIVEN VALUE`; the same joint model remains loaded. This separates the global multiplier from the conditional normalization instead of making you rename variables.

(c) Derive the conditional density.

$$\boxed{f_{X|Y}(x \mid y) = \frac{f(x, y)}{f_Y(y)} = \frac{2x + y}{1 + y}, \quad 0 < x < 1, \quad 0 < y < 2.}$$

For each allowed y, it is zero outside $0 < x < 1$. The denominator belongs to the given variable Y.

In the joint-slice tool, choose `PDF AT POINT` and enter remaining value $x = 1/2$. It reports conditional density approximately 1 and joint density approximately $1/2$, agreeing with $(2x + 1)/2$ and $(2x + 1)/4$. Choose `INTERVAL PROB` with event bounds 0 and $1/2$ to obtain approximately $3/8$. These numerical checks are optional; the homework's requested density is the function boxed above.

Why this is easier. The app handles normalization and conditional arithmetic. You still derive the requested symbolic functions from the short integrals above.

8.3 Common Question 3: a device that fails when either component fails

The joint PDF is $(x+y)/8$ on the square $(0,2)^2$. The device lifetime is $T = \min(X, Y)$, so the requested event is

$$\{T \le 1\} = \{X \le 1\} \cup \{Y \le 1\}.$$

Its complement is the single rectangle $X > 1, Y > 1$.

Quick load: choose `Q3 SERIES LIFETIME`, continue past the survival/complement reminder, and enter $N = 4$.

Manual route. Choose `DENSITY + INTEGRAL > DOUBLE INTEGRAL`:

Input	Value
Joint integrand `q(X,Y)`	`(X+Y)/8`
Fixed parameter `A`	`0`
Outer x limits	`1, 2`
Inner y limits	`1, 2`
Multiplier `g(X,Y)`	`1`
Outer resolution `N`	`4`

Read `RAW I`, approximately 3/8, on both resolution screens. This is the survival probability because you entered the already normalized joint density over the survival rectangle. Do *not* choose `NORMALIZE FULL S`: the entered rectangle is an event region, not the original support.

Subtract the survival probability from one:

$$\boxed{P(T \le 1) = 1 - \frac{3}{8} = \frac{5}{8} = 0.625.}$$

You can do $1 - 3/8$ on the home screen after leaving the solver. The preset also reminds you that failure is `1-RAW I`; the solver itself computes the survival integral.

What to write.

$$P(T \le 1) = 1 - P(X > 1, Y > 1) = 1 - \int_1^2 \int_1^2 \frac{x+y}{8}\, dy\, dx = \frac{5}{8}.$$

The event translation earns the reasoning; the app handles the integration and arithmetic. Do not multiply marginal survival probabilities, because independence was not given.

Why this is easier. A complement turns an L-shaped union into one rectangle. The app then computes its double integral in one run, with no separate intersection correction and no assumed independence.

8.4 STA 431 Question 4: two discrete event probabilities

The PMF is $p(x, y) = (x + y)/32$ for $x \in \{1, 2\}$ and $y \in \{1, 2, 3, 4\}$.

Quick load: choose `Q4 PMF EVENTS`; both event queries use the same loaded table.

Manual alternative: load once. Choose `DISCRETE + JOINT > FORMULA/GRID`. Enter x list `{1,2}`, y list `{1,2,3,4}`, and weight formula `X+Y`. The weight total is 32, so the app reconstructs the stated PMF automatically.

(a) $P(X > Y)$**.** From `TABLE TOOLS`, choose `EVENT PROB > P(EVENT)`. At `EVENT X,Y?`, enter `X>Y`. The app returns

$$\boxed{P(X > Y) = \frac{3}{32} = 0.09375.}$$

For the written solution, only the pair $(2, 1)$ satisfies the event, giving $p(2, 1) = (2 + 1)/32$.

(b) $P(X + Y \leq 3)$**.** You return to `EVENT TYPE`, so choose `P(EVENT)` again and enter `X+Y<=3`, using the actual $\leq$ token rather than separate `<` and `=` characters. The result is

$$\boxed{P(X + Y \leq 3) = \frac{1}{4} = 0.25.}$$

The contributing pairs are $(1, 1)$, $(1, 2)$, and $(2, 1)$, and

$$P(X + Y \leq 3) = \frac{2 + 3 + 3}{32} = \frac{8}{32} = \frac{1}{4}.$$

Why this is easier. You enter the support and weight formula once, then change only the event expression. The app selects the right pairs and adds their unequal weights. Counting qualifying pairs and dividing by eight would be wrong because the pairs are not equally likely.

A useful way to check your entry. The support is discrete. In part (a), Y cannot be 1.5; only the listed integer pairs matter. In part (b), the boundary pairs with sum exactly 3 must be included.

8.5 STA 431 Question 5: $P(X < Y)$ from a joint PDF

Use $f(x, y) = (2x + y)/4$ on $0 < x < 1$, $0 < y < 2$.

Build the event region. For a fixed $x \in (0, 1)$, the event $x < y$ and the support $y < 2$ give $x < y < 2$. Thus

$$P(X < Y) = \int_0^1 \int_x^2 \frac{2x + y}{4} \, dy \, dx.$$

The lower inner limit is x, not 0. Once that support intersection is correct, the calculator can do the entire numerical integration.

Quick load: choose `Q5 X LESS Y`, then enter $N = 4$.

Manual route. Choose `DENSITY + INTEGRAL > DOUBLE INTEGRAL`:

Input	Value
`q(X,Y)?`	`(2X+Y)/4`
`FIXED A?`	`0`
Outer x limits	`0, 1`
Inner y limits	`X, 2`
`g(X,Y),1=AREA?`	`1`
`EVEN N 4-100?`	`4`

Both resolution screens give `RAW I` approximately $19/24 = 0.7916666667$. Use this raw value as the probability. Normalizing the event region would answer a different question.

Exact written calculation.

$$\begin{aligned} P(X < Y) &= \int_0^1 \left[x + \frac{1}{2} - \frac{5}{8}x^2 \right] dx \\ &= \frac{1}{2} + \frac{1}{2} - \frac{5}{24} = \boxed{\frac{19}{24}}. \end{aligned}$$

The app removes both numerical integrations; writing the first iterated integral shows that you understood the event. The simplified single integral establishes the exact fraction.

Why the chosen order is easier. Integrating first in y needs one region. Reversing the order forces a split at $y = 1$:

$$\int_0^1 \int_0^y f(x, y) \, dx \, dy + \int_1^2 \int_0^1 f(x, y) \, dx \, dy.$$

Both are correct, but the first setup requires one app run and fewer boundary decisions.

8.6 Optional STA 631 Question 4: triangular support

Here $f(x,y) = 24x(1-x-y)$ on $x>0, y>0, x+y<1$. This is extra practice; it is separate from the STA 431 Question 4.

Quick load: from `LOAD + SOLVE > OPTIONAL 631`, choose `HW4 Q4 TRIANGLE` and enter $N = 4$.

Manual normalization check. Use `DOUBLE INTEGRAL` with `q=24X*(1-X-Y)`, $A = 0$, outer x limits $0, 1$, inner y limits 0, `1-X`, multiplier 1, and $N = 4$. The raw integral is approximately 1, confirming the proposed normalization numerically. The sloping upper bound is the key entry.

(a) Marginal formulas require the appropriate slices.

$$\boxed{f_X(x) = \int_0^{1-x} 24x(1-x-y)\,dy = 12x(1-x)^2, \quad 0<x<1,}$$

$$\boxed{f_Y(y) = \int_0^{1-y} 24x(1-x-y)\,dx = 4(1-y)^3, \quad 0<y<1.}$$

Both densities are zero otherwise. Each upper limit comes from $x+y<1$.

Numerical checks of the two marginals, with original variable names. Choose `DENSITY + INTEGRAL > JOINT SLICE`. Use joint shape `24X*(1-X-Y)` and known $c = 1$ in both directions:

Target	Direction, fixed value, remaining-variable bounds	Marginal output
$f_X(1/2)$	`Y GIVEN X`; given $1/2$; bounds 0, `1-X`	$3/2$
$f_Y(1/2)$	`X GIVEN Y`; given $1/2$; bounds 0, `1-Y`	$1/2$

The first result exceeds one because it is a density value, not an interval probability. The app keeps the fixed coordinate in place while integrating the remaining one. These spot checks verify arithmetic at chosen values; the integral formulas above supply the complete marginal densities.

(b) Independence. The marginals are positive throughout $(0,1)$, but the joint PDF is zero wherever $x+y>1$. For example, on the positive-area rectangle $0.6<x<0.8$, $0.6<y<0.8$, the marginal product is positive while the joint PDF is zero. Therefore $f_{X,Y} \neq f_X f_Y$ on a set of positive area and the variables are dependent.

Why the app helps. It checks the normalization and evaluates marginal slices with a moving bound. The app cannot replace the symbolic marginal derivations or the support-based independence argument.

8.7 Optional STA 631 Question 5: conditional reimbursement

The joint density is $f(x,y) = (20-x)/(25x)$ on $10 < x < 20$, $x/2 < y < x$. Here X is spending and Y is reimbursement.

(a) Marginal of spending. The density is constant in y over an interval of length $x/2$, so

$$\boxed{f_X(x) = \int_{x/2}^{x} \frac{20-x}{25x}\,\mathrm{d}y = \frac{20-x}{50}, \quad 10 < x < 20,}$$

with zero elsewhere. The interval length is more efficient than doing a long integration.

App check of (a). Choose `DENSITY + INTEGRAL > JOINT SLICE > Y GIVEN X`. Enter `q=(20-X)/(25X)`, known $c = 1$, given spending $X = 12$, lower reimbursement bound `X/2`, and upper bound `X`. The app reports marginal density $f_X(12) \approx 4/25$. Original X remains the spending variable; the app integrates in Y. It also reports conditional mean 9 and variance 3, which are useful checks but are not required here.

(b) Conditional density of reimbursement.

$$\boxed{f_{Y|X}(y \mid x) = \frac{(20-x)/(25x)}{(20-x)/50} = \frac{2}{x}, \quad \frac{x}{2} < y < x, \quad 10 < x < 20.}$$

Thus $Y \mid X = x$ is uniform on $(x/2, x)$. At $x = 12$ its density is $1/6$ on $(6, 12)$.

(c) Continue in the same joint-slice model. Choose `INTERVAL PROB`, enter event lower 8 and event upper 12, and read approximately

$$\boxed{P(Y \geq 8 \mid X = 12) = \frac{2}{3}.}$$

No density re-entry or variable renaming is needed. Alternatively, the quick-load route `OPTIONAL 631 > HW4 Q5 REIMBURSE` preloads the equivalent uniform slice on $(6, 12)$; choose its `INTERVAL PROB` and enter 8,12. Write the exact calculation as

$$P(Y \geq 8 \mid X = 12) = \int_{8}^{12} \frac{1}{6}\,\mathrm{d}y = \frac{12-8}{12-6} = \frac{2}{3}.$$

Why this is easier. Recognizing the uniform conditional law reduces the problem to an interval-length ratio. The app automatically divides the event integral by the full conditional-slice mass. You never divide by $P(X = 12)$, which is zero in this continuous model.

9 Homework 5: solve every problem with STAM2

These instructions cover Common Problems 1–7, the STA 431 Problem 8, and the separately marked STA 631 Problem 8. Start with `PRGM > STAM2`. The paths below begin at its main menu. Press `ENTER` to advance through result pages and use `BACK` to return through menus. For the shortest route, use `HOMEWORK MAP > LOAD + SOLVE` and select the matching question. The loader supplies inputs to the same general solvers; it does not merely display an answer. The manual routes below let you reproduce each calculation with changed data.

Use the calculator's actual X, Y, $e^{(}$, square, and comparison tokens. Enter a negative number with the unary `(-)` key. Fractions such as `1/4` can be entered directly. Fraction-looking outputs from numerical integration are convenient approximations; the displayed integral and its algebra establish the exact result.

9.1 Problem 1: normalize a PMF, then find its mean and variance

Quick load: group `HW5 Q1 TO Q4`, then `Q1 PMF MOMENTS`. Choose `MOMENTS` after normalization.

The model is $p_X(x) = c(x^2 + x)$ for $x = 1, 2, 3, 4$.

App route. Select `DISCRETE + JOINT > FORMULA/GRID`.

Prompt	Enter
`x VALUES LIST?`	`{1,2,3,4}`
`y VALUES LIST?`	`{0}`; this dummy coordinate makes a one-variable grid.
`q(X,Y)?`	`X^2+X`

The program generates the four weights $2, 6, 12, 20$, reports total weight 40, and displays $c = 1/40$. Choose `MOMENTS`. The X result page gives

$$\boxed{\mathbb{E}X = \frac{13}{4}, \qquad \mathbb{E}X^2 = \frac{227}{20}, \qquad \operatorname{Var} X = \frac{63}{80}.}$$

The app now recognizes the one-variable grid and suppresses dummy Y and correlation pages.

Write for parts (a)–(c).

$$1 = c(2 + 6 + 12 + 20) = 40c \quad \Rightarrow \quad c = \frac{1}{40},$$

$$\mathbb{E}X = \frac{2 + 12 + 36 + 80}{40} = \frac{13}{4}, \qquad \mathbb{E}X^2 = \frac{2 + 24 + 108 + 320}{40} = \frac{227}{20},$$

$$\operatorname{Var} X = \mathbb{E}X^2 - (\mathbb{E}X)^2 = \frac{227}{20} - \frac{169}{16} = \frac{63}{80}.$$

Why the app helps. One formula generates the values, sums the weights, normalizes them, and computes both moments and the variance. You do not need to enter four separate probabilities or repeat the weighted arithmetic.

9.2 Problem 2: a symmetric continuous variable and its square

Quick load: group `HW5 Q1 TO Q4`, then `Q2 PDF AND SQUARE`. Choose `MEAN+VARIANCE`, then `FUNCTION g(X)` with `X^2`.

The given density is $f_X(x) = \frac{3}{4}(1 - x^2)$ on $(-1, 1)$. Both X and $Y = X^2$ are involved.

App route. Select `DENSITY + INTEGRAL > 1D PDF OR SLICE`.

Prompt	Enter
`q(X)?`	`1-X^2`
`FIXED A?`	`0`; this model has no parameter A.
`LOWER X?`, `UPPER X?`	`(-)1`, then `1`

This manual route omits the known factor 3/4 because the tool normalizes its shape. It reports $\int q = 4/3$ and normalizing factor 3/4. A preset that retains the given normalized density instead reports total mass 1; both routes yield the same moments.

Choose `MEAN+VARIANCE`. For part (a), read

$$\boxed{\mathbb{E}X = 0, \qquad \mathbb{E}X^2 = \frac{1}{5}, \qquad \operatorname{Var} X = \frac{1}{5}.}$$

Then choose `FUNCTION g(X)` and enter `X^2`. The output labels now refer to $g(X) = Y$, so part (b) is

$$\boxed{\mathbb{E}Y = \frac{1}{5}, \qquad \mathbb{E}Y^2 = \frac{3}{35}, \qquad \operatorname{Var} Y = \frac{8}{175}.}$$

The program computes the centered integral for the transformed variance; you do not have to derive the distribution of X^2.

Write the mathematical setup. Symmetry makes $xf_X(x)$ odd, hence $\mathbb{E}X = 0$. For the even moments,

$$\mathbb{E}X^2 = \frac{3}{2}\int_0^1 (x^2 - x^4)\,\mathrm{d}x = \frac{1}{5}, \qquad \mathbb{E}X^4 = \frac{3}{2}\int_0^1 (x^4 - x^6)\,\mathrm{d}x = \frac{3}{35}.$$

Then $\mathbb{E}Y = \mathbb{E}X^2$ and $\operatorname{Var} Y = \mathbb{E}X^4 - (\mathbb{E}X^2)^2 = 3/35 - 1/25 = 8/175$.

Why the app helps. Enter the density once, then change only the transformation to X^2. The same solver performs all the required finite integrals and uses the fourth moment automatically when calculating $\operatorname{Var}(X^2)$.

9.3 Problem 3: derive an exponential MGF and obtain two moments

Quick load: group `HW5 Q1 TO Q4`, then `Q3 EXPONENTIAL MGF`. Advance through the moments and MGF pages.

Here $f_X(x) = 3e^{-3x}$ for $x > 0$, so the exponential *rate* is 3.

App route. Select `MGF TOOLS > EXPONENTIAL RATE`.

Prompt	Enter
`RATE L>0?`	`3`
`SCALE a?`	`1`
`SHIFT b?`	`0`

Scale 1 and shift 0 request the original variable. Advance through the output pages. They give

$$\boxed{M_X(t) = \frac{3}{3 - t}, \quad t < 3; \qquad \mathbb{E}X = \frac{1}{3}, \quad \mathbb{E}X^2 = \frac{2}{9}.}$$

The tool also reports variance 1/9. Its general transformed-MGF display simplifies to the original expression when $a = 1, b = 0$.

Write part (a), including convergence.

$$M_X(t) = \mathbb{E}(e^{tX}) = \int_0^\infty 3e^{-(3-t)x}\,\mathrm{d}x = \frac{3}{3-t} \quad \text{for } t < 3.$$

The improper integral converges precisely when $3 - t > 0$. A finite numerical cutoff would not establish this domain; the exponential template records the relevant analytic result.

Write part (b).

$$M_X'(t) = \frac{3}{(3-t)^2}, \qquad M_X''(t) = \frac{6}{(3-t)^3},$$

so $M_X'(0) = 1/3$ and $M_X''(0) = 2/9$.

Why the app helps. The exact exponential template supplies the MGF form, its domain, and both moment values from one parameter. Your written work then shows the short integral and derivative argument required by the question.

9.4 Problem 4: center and scale a given MGF

Quick load: group `HW5 Q1 TO Q4`, then `Q4 AFFINE MGF`. The original MGF and transformation are already supplied.

The MGF is $M_X(t) = e^{4t+8t^2}$ and $Z = (X-4)/4 = \frac{1}{4}X - 1$.

App route. Select `MGF TOOLS > QUADRATIC LOG MGF`.

Prompt	Enter
`C?`	`4`; coefficient of t in $\log M_X(t)$.
`D` ≥ 0`?`	`8`; coefficient of t^2, equal to half the original variance.
`SCALE a?`	`1/4`
`SHIFT b?`	`(-)1`

The original moments page gives mean 4, second moment 32, and variance 16. The transformed page gives mean 0, second moment 1, and variance 1. On `TRANSFORMED MGF`, the form is e^{Ft+Ht^2} with $F = 0$ and $H = 1/2$:

$$\boxed{M_Z(t) = e^{t^2/2}, \quad \mathbb{E}Z = 0, \quad \operatorname{Var} Z = 1.}$$

Both MGFs are finite for every real t.

Write part (a). Use the affine rule before simplifying:

$$M_Z(t) = e^{-t}M_X(t/4) = e^{-t}\exp\left(4\frac{t}{4} + 8\frac{t^2}{16}\right) = e^{t^2/2}.$$

For part (b), $M_Z'(t) = te^{t^2/2}$ and $M_Z''(t) = (1+t^2)e^{t^2/2}$, so $\mathbb{E}Z = 0$, $\mathbb{E}Z^2 = 1$, and $\operatorname{Var} Z = 1$.

Why the app helps. It keeps track of the shift, the substitution into the old MGF, and the factor of two between a quadratic log-MGF coefficient and a variance. This problem matches an exact template, so numerical differentiation is unnecessary.

9.5 Problem 5: variance of the updated insurance benefit

Quick load: group `HW5 Q5 TO Q8`, then `Q5 CLAIM VARIANCE`. Advance to the computed variance.

First express the new total as

$$T = W + V = 2X + 1.1Y + 100.$$

Only variances and covariance are supplied; a mean is not needed.

App route. Select `LINEAR COMBOS > ONE COMBINATION > VAR/COV ONLY`. Enter values in the order shown:

Prompt	Enter
`n (1-3)?`	`2`
Variance of $X_1 = X$, then $X_2 = Y$	`300`, then `500`
`COV(X1,X2)`	`150`
`a FOR U?` for X_1, then X_2	`2`, then `1.1` (or `11/10`)
`OFFSET c FOR U?`	`100`

The result is $\boxed{\mathrm{Var}(T) = 2465}$. The display correctly says `MEAN NOT PROVIDED`; do not fill in an invented mean.

Write the variance expansion.

$$\begin{aligned}\mathrm{Var}(T) &= 4\,\mathrm{Var}\,X + (1.1)^2\,\mathrm{Var}\,Y + 2(2)(1.1)\,\mathrm{Cov}(X,Y)\\ &= 4(300) + 1.21(500) + 4.4(150)\\ &= 1200 + 605 + 660 = 2465.\end{aligned}$$

The fixed increase of 100 contributes no variance.

Why the app helps. Enter the coefficient and covariance data once; it forms all square and cross terms. The variance-only mode avoids asking for information the problem does not supply.

9.6 Problem 6: a signed combination of three variables

Quick load: group `HW5 Q5 TO Q8`, then `Q6 THREE VARIABLES`. All supplied variances, covariances, and coefficients are prefilled.

The target is $W = 3X - 2Y + Z$.

App route. Select `LINEAR COMBOS > ONE COMBINATION > VAR/COV ONLY`.

Input group	Enter in this order
Number of variables	`3`
Variances of $X_1 = X$, $X_2 = Y$, $X_3 = Z$	`4`, `9`, `2`
Covariances 12, 13, 23	`1`, `(-)1`, `0`
Coefficients for X_1, X_2, X_3	`3`, `(-)2`, `1`
Offset	`0`

The result page gives $\boxed{\mathrm{Var}(W) = 56}$.

Write all three covariance terms.

$$\begin{aligned}\operatorname{Var}(W) &= 9\operatorname{Var}X + 4\operatorname{Var}Y + \operatorname{Var}Z \\ &\quad - 12\operatorname{Cov}(X,Y) + 6\operatorname{Cov}(X,Z) - 4\operatorname{Cov}(Y,Z) \\ &= 36 + 36 + 2 - 12 - 6 - 0 = 56.\end{aligned}$$

The coefficient -2 changes two cross-term signs. The negative covariance $\operatorname{Cov}(X,Z) = -1$ then makes $6\operatorname{Cov}(X,Z) = -6$.

Why the app helps. Three variances and three distinct covariances determine the calculation. It assembles every pair once with the correct signed coefficient products, reducing the chance of omitting a term or its factor of two.

9.7 Problem 7: marginal and conditional densities, then conditional moments

Quick load: group `HW5 Q5 TO Q8`, then `Q7 CONDITIONAL`. This loads the slice at $Y = 1/2$; select `MEAN+VARIANCE`.

The joint PDF is $f(x,y) = x + y$ on the unit square.

Parts (a)–(b): write the short symbolic calculation.

$$f_Y(y) = \int_0^1 (x+y)\,\mathrm{d}x = \frac{1}{2} + y, \qquad 0 < y < 1,$$

with zero density elsewhere. Therefore

$$f_{X|Y}(x \mid y) = \frac{x+y}{1/2+y}, \qquad 0 < x < 1, \quad 0 < y < 1,$$

and the conditional density is zero outside its x slice. The new `JOINT SLICE` tool now checks marginal and conditional density values directly. Select `DENSITY + INTEGRAL > JOINT SLICE > X GIVEN Y`, enter `q(X,Y)=X+Y`, known multiplier `1`, given value `1/2`, and remaining-variable bounds `0`, `1`. It reports the marginal density $f_Y(1/2) = 1$ alongside the conditional moments below. In `SLICE OPTIONS`, `NEW GIVEN VALUE` lets you change y without retyping the model; `PDF AT POINT` evaluates the conditional and joint densities at a remaining x value. For example, at $y = 1/4$ the marginal is $3/4$, and at $x = 1/4$ the conditional density is $2/3$. These are density values, not probabilities at isolated points. A numerical calculation verifies values of these functions; it does not produce a symbolic formula in y.

Part (c): app route. Select `DENSITY + INTEGRAL > 1D PDF OR SLICE`.

Prompt	Enter
`q(X)?`	`X+A`
`FIXED A?`	`1/2`; A is the given value of Y.
`LOWER X?`, `UPPER X?`	`0`, then `1`

The slice mass is $D = f_Y(1/2) = 1$. Select `MEAN+VARIANCE`. Read

$$\boxed{\mathbb{E}(X \mid Y = 1/2) = \frac{7}{12}, \qquad \mathbb{E}(X^2 \mid Y = 1/2) = \frac{5}{12}, \qquad \operatorname{Var}(X \mid Y = 1/2) = \frac{11}{144}.}$$

For your written calculation,

$$\int_0^1 x(x+1/2)\,\mathrm{d}x = \frac{7}{12}, \qquad \int_0^1 x^2(x+1/2)\,\mathrm{d}x = \frac{5}{12},$$

then $5/12 - (7/12)^2 = 11/144$.

Why the app helps. Enter the unnormalized joint slice $X + A$ directly. The solver computes its mass and divides by it, so conditional normalization and both moment integrals are handled together. This also works at other valid given values of Y after updating the parameter and recomputing the slice.

9.8 STA 431 Problem 8: four tasks from one joint table

Quick load: group `HW5 Q5 TO Q8`, then `Q8 JOINT MOMENTS`. Choose `MOMENTS`; keep this dataset for the conditional part.

The PMF is $(x + y)/36$ on $x, y \in \{1, 2, 3\}$.

For both complete marginal tables, choose `MARG/CONDITIONAL > MARGINAL TABLES`.

Load once. Select `DISCRETE + JOINT > FORMULA/GRID`; enter `{1,2,3}` for each coordinate list and `X+Y` for the weight formula. The app generates nine rows and reports total 36 and normalizer 1/36.

Parts (a)–(c). Choose `MOMENTS` and advance through the X, Y, and product pages:

$$\boxed{\mathbb{E}X = \mathbb{E}Y = \frac{13}{6}, \qquad \mathbb{E}XY = \frac{14}{3}, \qquad \mathrm{Cov}(X, Y) = -\frac{1}{36}.}$$

The additional marginal variances are both 23/36, and correlation is $-1/23$; these are useful checks but are not requested by this question.

Part (d). Return to `TABLE TOOLS`, choose `MARG/CONDITIONAL > Y GIVEN X`, and enter `1`. The conditional-distribution page gives

y	1	2	3
$P(Y = y \mid X = 1)$	2/9	3/9	4/9

and the next page gives $\boxed{\mathbb{E}(Y \mid X = 1) = 20/9}$. These conditional weights sum to one.

Write the compact derivation. The marginals are

$$p_X(x) = \frac{x + 2}{12}, \quad p_Y(y) = \frac{y + 2}{12} \qquad (x, y = 1, 2, 3).$$

Thus $\mathbb{E}X = (3 + 8 + 15)/12 = 13/6$, with the same result for Y. Since $\sum_{k=1}^{3} k = 6$ and $\sum_{k=1}^{3} k^2 = 14$,

$$\mathbb{E}XY = \frac{14(6) + 6(14)}{36} = \frac{14}{3}, \qquad \mathrm{Cov}(X, Y) = \frac{14}{3} - \left(\frac{13}{6}\right)^2 = -\frac{1}{36}.$$

Finally $p_X(1) = 1/4$, so

$$p_{Y|X}(y \mid 1) = \frac{(1 + y)/36}{1/4} = \frac{1 + y}{9}, \quad y = 1, 2, 3,$$

with zero mass elsewhere, and $\mathbb{E}(Y \mid X = 1) = (2 + 6 + 12)/9 = 20/9$.

Why the app helps. A single formula-generated table serves all four parts. It computes both marginal moments, the product moment, covariance, the full requested conditional PMF, and its mean without re-entering the model.

9.9 Optional STA 631 Problem 8: zero covariance without independence

Quick load: group `OPTIONAL 631`, then `HW5 Q8 ZERO COV`. Choose moments, functions, or interval probability as described below.

This problem uses $f_X(x) = 1 - |x|$ on $(-1, 1)$ and $Y = X^2$. It is also a useful counterexample for the STA 431 independence topic.

App calculations. Select `DENSITY + INTEGRAL > 1D PDF OR SLICE`; enter `1-abs(X)`, fixed parameter `0`, and limits `(-)1`, `1`. The mass is 1. `MEAN+VARIANCE` gives $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1/6$. In `FUNCTION g(X)`, enter `X^3` to obtain $\mathbb{E}X^3 = 0$.

Part (a): the proof. Symmetry makes the first and third raw moments zero. Because $Y = X^2$,

$$\mathrm{Cov}(X, Y) = \mathbb{E}X^3 - \mathbb{E}X\mathbb{E}X^2 = 0 - 0 \cdot \frac{1}{6} = 0.$$

The symmetry argument supplies exact zeros; numerical output checks the moment calculation.

Part (b): build explicit events. Let

$$A = \{-1/2 < X < 1/2\}, \qquad B = \{0 \le Y < 1/4\}.$$

Because $Y = X^2$, the events coincide up to zero-probability endpoints. Use `INTERVAL PROB` with lower `(-)1/2` and upper `1/2`; the app gives

$$P(A) = \int_{-1/2}^{1/2} (1 - |x|)\,\mathrm{d}x = \frac{3}{4}.$$

Thus $P(B) = P(A \cap B) = 3/4$, but

$$\boxed{P(A \cap B) = \frac{3}{4} \neq \frac{9}{16} = P(A)P(B).}$$

The variables are dependent despite zero covariance.

Why the app helps. It evaluates the moment and interval integrals. The dependence proof comes from relating the two events through $Y = X^2$. Use the supplied one-variable density: the pair (X, X^2) lies on a curve and does not have an ordinary two-dimensional area density for the double-integral solver.

10 Homework-style practice: 70 questions

These are original practice problems modeled on the skills in Homeworks 4 and 5, not predictions of the examination. Each subsection contains ten questions. Unless specified otherwise, a density or mass function is zero outside its stated support. Give exact answers and show the probability model, support, sum or integral, and final calculation. Solutions are collected after the complete question bank.

10.1 A. Multivariate distributions

A1. Let $p(x, y) = c(x + 2y)$ for $x \in \{1, 2\}$ and $y \in \{0, 1, 2\}$. Find c, both marginal PMFs, and $p_{X|Y}(x \mid 1)$. Determine whether X and Y are independent.

A2. Let $f(x, y) = c(x + 2y)$ for $0 < x < 2$, $0 < y < 1$. Find c, f_X, f_Y, and $f_{X|Y}(x \mid y)$, including each support.

A3. The joint density is $f(x, y) = 2$ on $0 < y < x < 1$. Sketch the support, derive both marginals, and calculate $P(Y < X/2)$. Are the variables independent?

A4. A system works only while both components work. Their lifetimes have density $f(x, y) = (x + y)/27$ for $0 < x < 3$, $0 < y < 3$. Translate "the system fails during the first hour" into an event and find its probability. Do not assume independent lifetimes.

A5. Let $p(x, y) = (x+y)/21$ for $x \in \{1, 2\}$, $y \in \{1, 2, 3\}$. List the contributing pairs and find (a) $P(X > Y)$ and (b) $P(X + Y \le 4)$.

A6. Let $f(x, y) = \frac{2}{3}(2x + y)$ on the unit square. Find $P(X < Y)$ using a single iterated integral. Give a second setup with the integration order reversed.

A7. Define $A(x) = 0$ for $x \le 0$, $A(x) = x^2$ for $0 < x < 1$, and $A(x) = 1$ for $x \ge 1$. Define $B(y) = 0$ for $y \le 0$, $B(y) = y/2$ for $0 < y < 2$, and $B(y) = 1$ for $y \ge 2$. A joint CDF is $F(x, y) = A(x)B(y)$. Find the joint density, both marginals, independence, and $P(1/2 < X \le 1,\ 1/2 < Y \le 3/2)$.

A8. Suppose $f_X(x) = 2x$ for $0 < x < 1$, and $Y \mid X = x$ is uniform on $(0, x)$. Reconstruct $f_{X,Y}$ and derive f_Y. Explain why the conditional support matters.

A9. Both X and Y are Bernoulli with $P(X = 1) = P(Y = 1) = 1/2$. Construct two different joint PMF tables with these same marginals: one independent and one dependent. Find $P(X = Y)$ in each case. Can the joint distribution be recovered from the marginals alone?

A10. *Bounds challenge.* Let $f(x, y) = c$ for $0 < x < 1$ and $x^2 < y < 1$. Find c, f_Y, and $P(Y < 1/4)$. Write the probability integral in both orders.

10.2 B. Expectation, LOTUS, and properties

B1. Let $p_X(x) = c(x^2 + 1)$ for $x \in \{0, 1, 2, 3\}$. Find c, $\mathbb{E}[X]$, and $\mathbb{E}[2X + 3]$.

B2. Let $f_X(x) = 3x^2$ for $0 < x < 1$. Calculate $\mathbb{E}[X]$ and $\mathbb{E}[1/X]$ directly from the density. Explain why $\mathbb{E}[1/X]$ is not generally $1/\mathbb{E}[X]$.

B3. Let $f_X(x) = 1 - |x|$ for $-1 < x < 1$. Use symmetry and integration to obtain $\mathbb{E}[X]$, $\mathbb{E}[|X|]$, and $\mathbb{E}[X^2]$.

B4. A variable X takes values $0, 1, 2$ with probabilities $1/4, 1/2, 1/4$. For $Y = X^2$, find $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$ without first deriving a distribution formula for Y.

B5. If $f_X(x) = 2x$ on $(0, 1)$, calculate $\mathbb{E}[(1 - X)^2]$ and $\mathbb{E}[\log X]$. State the version of LOTUS being used.

B6. A company receives a payment of \$100 and pays \$20 per service call. The number X of calls is $0, 1, 3$ with probabilities $1/2, 3/10, 1/5$. Express profit as a random variable and find expected profit.

B7. The joint PMF table is

	$Y = 0$	$Y = 2$
$X = 0$	$1/10$	$2/10$
$X = 1$	$3/10$	$4/10$

Calculate $\mathbb{E}[3X - 2Y + 5]$ in two ways: directly from the joint table and by linearity.

B8. You know only that $\mathbb{E}[X] = -2$ and $\mathbb{E}[Y] = 4$, with finite expectations. Find $\mathbb{E}[3X - \frac{1}{2}Y + 7]$. Is independence necessary? Is there enough information to find $\mathbb{E}[XY]$?

B9. *Existence check.* Let $f_X(x) = x^{-2}$ for $x \geq 1$. Verify normalization, determine whether $\mathbb{E}[X]$ is finite, and calculate $\mathbb{E}[1/X]$.

B10. For $f_X(x) = 2x$, $0 < x < 1$, let $I = \mathbf{1}_{\{X > 1/2\}}$. Calculate $\mathbb{E}[I]$ and $\mathbb{E}[XI]$. Interpret the first expectation as a probability.

10.3 C. Raw moments, central moments, and variance

C1. Let $p_X(x) = x/6$ for $x = 1, 2, 3$. Find the first three moments about the origin, the variance, and the third moment about the mean.

C2. Let $f_X(x) = 4x^3$ for $0 < x < 1$. Derive a formula for $\mathbb{E}[X^r]$ for nonnegative integers r, then find the mean, variance, and third central moment.

C3. Suppose the first four raw moments are $2, 7, 26, 107$, respectively. Find the variance and the third and fourth central moments by expanding powers of $X - \mu$.

C4. If $\mathbb{E}[X] = 2$ and $\mathrm{Var}(X) = 5$, let $Y = 3X - 4$. Find $\mathbb{E}[X^2]$, $\mathbb{E}[Y]$, $\mathrm{Var}(Y)$, and $\mathbb{E}[Y^2]$.

C5. Let X be uniform on $(-2, 2)$ and $Y = X^2$. Find $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$ by integrating powers of X.

C6. Let $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 > 0$. Prove that $Z = (X - \mu)/\sigma$ has mean zero and variance one. Express $\mathbb{E}[X^2]$ in terms of μ and σ.

C7. *Existence check.* Let $f_X(x) = 2x^{-3}$ for $x \geq 1$. Find $\mathbb{E}[X]$ and determine whether the variance is finite. Identify the integral responsible for your conclusion.

C8. Let X be uniform on $\{-2, -1, 0, 1, 2\}$. Find its mean, variance, fourth central moment, and standardized fourth moment $\mathbb{E}[(X - \mu)^4]/\sigma^4$.

C9. A random variable has mean 3 and variance 4. Find $\mathbb{E}[(X-5)^2]$. Explain why this is not the variance of X.

C10. Let $f_X(x) = \frac{3}{2}x^2$ for $-1 < x < 1$ and $Y = X^2$. Find $\mathbb{E}[X]$, $\mathrm{Var}(X)$, $\mathbb{E}[Y]$, and $\mathrm{Var}(Y)$. Which moments of X are needed for the last calculation?

10.4 D. Moment generating functions

D1. For $f_X(x) = 5e^{-5x}$, $x > 0$, derive $M_X(t)$ and its domain. Use derivatives at zero to find $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathrm{Var}(X)$.

D2. A variable takes values $-1, 0, 2$ with probabilities $1/4, 1/2, 1/4$. Write its MGF and use it to calculate the mean and variance.

D3. Given $M_X(t) = \exp(6t + 9t^2/2)$, let $Z = (X-6)/3$. Derive $M_Z(t)$ and find its mean and variance.

D4. Suppose $M_X(t) = (1-2t)^{-3}$, $t < 1/2$. Find $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathrm{Var}(X)$ by differentiation. Then find the MGF of $X + 4$.

D5. Derive the MGF of a variable uniform on $(0, 2)$. Define its value at $t = 0$ and use its expansion near zero to obtain the first two raw moments.

D6. Independent variables X and Y have exponential densities with rates 2 and 4. Find the MGF, domain, mean, and variance of $S = 2X + Y$.

D7. Two variables have MGFs equal to $2/(2-t)$ for all t in an open interval containing zero. What can be concluded about their distributions? Find $P(X > 1)$ for either variable. Must the two variables be independent?

D8. If $M_X(t) = 1/(1-t)$ for $t < 1$, find the MGF and its domain for $W = 3 - 2X$. Find $\mathbb{E}[W]$ and $\mathrm{Var}(W)$.

D9. A proposed MGF is $M(t) = \exp(2t - t^2)$. Can it be an MGF on a neighborhood of zero? Use the variance identity to justify your answer.

D10. *Existence challenge.* If $f_X(x) = x^{-2}$ for $x \geq 1$, show that $\mathbb{E}[e^{tX}]$ is infinite for every $t > 0$. Is finiteness at $t = 0$ alone enough for the usual MGF existence and uniqueness theorem?

10.5 E. Product moments, covariance, and correlation

E1. The joint PMF is

	$Y=0$	$Y=1$
$X=0$	2/10	3/10
$X=1$	1/10	4/10

Find $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$, $\mathrm{Cov}(X,Y)$, and $\mathrm{Corr}(X,Y)$. Check independence separately.

E2. For $f(x,y) = 4xy$ on the unit square, find both marginals, prove independence, and calculate $\mathbb{E}[XY]$, covariance, and correlation.

E3. For $f(x,y) = 2$ on $0 < y < x < 1$, calculate $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$, covariance, and correlation.

E4. Let X be uniform on $(-1, 1)$ and $Y = X^2$. Prove that covariance is zero. Then disprove independence using the events $A = \{|X| < 1/2\}$ and $B = \{Y < 1/4\}$.

E5. You are given $\mathbb{E}[X] = 2$, $\mathbb{E}[Y] = -1$, $\mathbb{E}[XY] = -5$, $\mathrm{Var}(X) = 4$, and $\mathrm{Var}(Y) = 9$. Find covariance and correlation.

E6. Let $Y = 5 - 3X$, where $\mathrm{Var}(X) = 2$. Find $\mathrm{Cov}(X, Y)$, $\mathrm{Var}(Y)$, and correlation. Explain the sign.

E7. If $\mathrm{Cov}(X, Y) = -2$, find $\mathrm{Cov}(2X + 7, -4Y + 1)$. Explain what happens to the added constants.

E8. Independent variables satisfy $P(X = 1) = 2/5$, $P(X = 0) = 3/5$, and $f_Y(y) = 2e^{-2y}$ for $y > 0$. Calculate the product moments $\mathbb{E}[X^2Y]$ and $\mathbb{E}[XY^2]$.

E9. Can $\mathrm{Var}(X) = 1$, $\mathrm{Var}(Y) = 4$, and $\mathrm{Cov}(X, Y) = 3$ hold simultaneously? State the allowable covariance interval and justify it.

E10. Suppose $\mathrm{Corr}(X, Y) = -2/5$ and both variances are positive. Find $\mathrm{Corr}(-2X + 3, -5Y + 4)$ and $\mathrm{Corr}(2X + 3, -5Y + 4)$. Derive the rule rather than assuming that correlation is unchanged under every transformation.

10.6 F. Moments of one and two linear combinations

F1. Let $\mathbb{E}[X] = 3$, $\mathbb{E}[Y] = -2$, $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Cov}(X, Y) = 1$. For $U = 2X - 3Y + 5$, calculate mean and variance.

F2. An updated claim pays $W = 1.5X + 40$ and $V = 2Y - 10$. Given $\mathrm{Var}(X) = 100$, $\mathrm{Var}(Y) = 64$, and $\mathrm{Cov}(X, Y) = 20$, find $\mathrm{Var}(W + V)$.

F3. Let $\mathrm{Var}(X) = 1$, $\mathrm{Var}(Y) = 4$, $\mathrm{Var}(Z) = 9$, $\mathrm{Cov}(X, Y) = 1/2$, $\mathrm{Cov}(X, Z) = -1$, and $\mathrm{Cov}(Y, Z) = 1$. Find $\mathrm{Var}(2X - Y + 3Z)$, displaying all three covariance terms.

F4. Given $\mathrm{Var}(X) = 2$, $\mathrm{Var}(Y) = 5$, and $\mathrm{Cov}(X, Y) = -1$, define $U = X + 2Y$, $V = 3X - Y$. Find $\mathrm{Var}(U)$, $\mathrm{Var}(V)$, $\mathrm{Cov}(U, V)$, and $\mathrm{Corr}(U, V)$.

F5. Five independent variables each have mean 4 and variance 3. Find the mean and variance of their sum and of their average. Which answers would still hold if independence were replaced by pairwise zero covariance?

F6. Let X be uniform on $(-1, 1)$ and $Y = X^2$. Find $\mathrm{Var}(2X + Y)$. Explain why the covariance term vanishes even though the variables are dependent.

F7. You know $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Var}(X + Y) = 20$. Find $\mathrm{Cov}(X, Y)$ and $\mathrm{Var}(X - Y)$.

F8. Let $\mathrm{Var}(X) = 9$, $\mathrm{Var}(Y) = 4$, and $\mathrm{Cov}(X, Y) = 2$. Find the real number a that minimizes $\mathrm{Var}(X - aY)$ and find that minimum variance.

F9. A combined cost is $T = 0.4X + 0.6Y$, where the means are 100 and 80, variances are 25 and 16, and correlation is $1/2$. Find $\mathbb{E}[T]$ and $\mathrm{Var}(T)$. Convert correlation into covariance before substituting.

F10. Use the variance and covariance data in F3. Define $U = X + Y$ and $V = Y - Z$. Find the two variances, covariance, and correlation.

10.7 G. Conditional expectation and conditional variance

Questions G5, G7, and G10 use the supplementary tower and total-variance laws; they extend the direct conditional calculations in the homework.

G1. Let $p(x,y) = (x+y)/15$ for $x \in \{1,2\}$ and $y \in \{0,1,2\}$. Find $p_{Y|X}(y \mid 1)$, $\mathbb{E}[Y \mid X=1]$, and $\mathrm{Var}(Y \mid X=1)$.

G2. Let $f(x,y) = \frac{2}{3}(2x+y)$ on the unit square. Derive f_Y and $f_{X|Y}$. Calculate $\mathbb{E}[X \mid Y=1/2]$ and $\mathrm{Var}(X \mid Y=1/2)$.

G3. Let $f(x,y) = 2$ on $0 < y < x < 1$. Find $\mathbb{E}[Y \mid X=x]$ and $\mathrm{Var}(Y \mid X=x)$ as functions of x. Evaluate them at $x = 4/5$.

G4. If X and Y are independent and $f_X(x) = 2x$, $0 < x < 1$, find $\mathbb{E}[X \mid Y=y]$ and $\mathrm{Var}(X \mid Y=y)$ for conditioning values where the conditional law is defined.

G5. Let X be uniform on $(0,1)$ and let $Y \mid X=x$ be Bernoulli with success probability x. Find $\mathbb{E}[Y \mid X]$, $\mathrm{Var}(Y \mid X)$, $\mathbb{E}[Y]$, and $\mathrm{Var}(Y)$ using total expectation and total variance.

G6. For $f(x,y) = 2$ on $0 < y < x < 1$, derive $\mathbb{E}[X \mid Y=y]$. Then use $f_Y(y) = 2(1-y)$ and total expectation to recover $\mathbb{E}[X]$.

G7. Let Y be uniform on $(0,2)$ and let $X \mid Y=y$ be uniform on $(0,y)$. Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$ using conditional moments rather than a joint-density calculation.

G8. Let $f(x,y) = c$ on $0 < x < 1$, $0 < y < x^2$. Find c and the conditional density of Y given $X=x$. Compute $P(Y > 1/8 \mid X = 1/2)$.

G9. For $f(x,y) = 2$ on $0 < y < x < 1$, find $\mathbb{E}[Y \mid X > 1/2]$ and compare it with $\mathbb{E}[Y \mid X = 1/2]$. Explain the difference between conditioning on an event and conditioning at a value.

G10. A variable Y takes the values $1, 2$ with equal probability. Suppose $\mathbb{E}[X \mid Y] = 1 + 2Y$ and $\mathrm{Var}(X \mid Y) = 3Y$. Find $\mathbb{E}[X]$, $\mathrm{Var}(X)$, and $\mathbb{E}[X^2]$.

11 Worked solutions to the 70 practice questions

The solution labels match the question bank. Values of a PMF or PDF outside the support are zero. Conditional formulas require a positive conditioning marginal; continuous conditioning uses a conditional density, not division by $P(Y = y)$.

11.1 A. Multivariate distributions

A1. The total unnormalized mass is $9 + 12 = 21$, so $c = 1/21$. Summing in the other coordinate gives

$$p_X(x) = \frac{3x + 6}{21}, \quad x = 1, 2; \qquad p_Y(y) = \frac{3 + 4y}{21}, \quad y = 0, 1, 2.$$

Thus $p_{X|Y}(x \mid 1) = (x + 2)/7$, namely $3/7, 4/7$. Independence fails: $p(1, 0) = 1/21$, whereas $p_X(1)p_Y(0) = (3/7)(1/7) = 3/49$.

A2. Normalization gives $1 = c \int_0^1 \int_0^2 (x + 2y)\, \mathrm{d}x\, \mathrm{d}y = 4c$, hence $c = 1/4$. Consequently

$$f_X(x) = \frac{x + 1}{4}, \ 0 < x < 2; \qquad f_Y(y) = \frac{1}{2} + y, \ 0 < y < 1.$$

For $0 < y < 1$, $f_{X|Y}(x \mid y) = (x + 2y)/(2 + 4y)$ on $0 < x < 2$. Integrating it over x gives one.

A3. The support lies below $y = x$ in the unit square. Vertical and horizontal slices give $f_X(x) = \int_0^x 2\, \mathrm{d}y = 2x$ and $f_Y(y) = \int_y^1 2\, \mathrm{d}x = 2(1 - y)$, each on $(0, 1)$. Also

$$P(Y < X/2) = \int_0^1 \int_0^{x/2} 2\, \mathrm{d}y\, \mathrm{d}x = \int_0^1 x\, \mathrm{d}x = \frac{1}{2}.$$

They are dependent: the product of marginals is positive above the diagonal, where the joint density is zero.

A4. System lifetime is $T = \min(X, Y)$. Its failure event is $\{X \le 1\} \cup \{Y \le 1\}$, up to endpoints of probability zero. Use the complement:

$$P(T \le 1) = 1 - \int_1^3 \int_1^3 \frac{x + y}{27}\, \mathrm{d}y\, \mathrm{d}x = 1 - \frac{16}{27} = \frac{11}{27}.$$

Multiplying marginal survival probabilities would require independence, which is not given.

A5. For (a), only $(2, 1)$ contributes, so the probability is $3/21 = 1/7$. For (b), the pairs are $(1, 1)$, $(1, 2)$, $(1, 3)$, $(2, 1)$, and $(2, 2)$, with weights $2, 3, 4, 3, 4$. Therefore $P(X + Y \le 4) = 16/21$.

A6. On the event region, $0 < x < y < 1$. Thus

$$P(X < Y) = \int_0^1 \int_0^y \frac{2}{3}(2x + y)\, \mathrm{d}x\, \mathrm{d}y = \frac{4}{3} \int_0^1 y^2\, \mathrm{d}y = \frac{4}{9}.$$

The reversed setup is $\int_0^1 \int_x^1 \frac{2}{3}(2x + y)\, \mathrm{d}y\, \mathrm{d}x$.

A7. Taking the mixed derivative inside the rectangle gives $f(x,y) = x$ for $0 < x < 1$, $0 < y < 2$. The marginal densities are $2x$ on $(0,1)$ and $1/2$ on $(0,2)$. Their product equals the joint density, so independence holds. Rectangle probabilities satisfy

$$[A(1) - A(1/2)][B(3/2) - B(1/2)] = (1 - 1/4)(3/4 - 1/4) = \frac{3}{8}.$$

The original CDF defines all boundary and exterior values through A and B.

A8. The conditional density is $f_{Y|X}(y \mid x) = 1/x$ for $0 < y < x$. Multiplication gives $f(x,y) = 2$ on $0 < y < x < 1$. Therefore $f_Y(y) = \int_y^1 2 \, \mathrm{d}x = 2(1-y)$, $0 < y < 1$. Omitting the restriction $y < x$ would incorrectly spread mass over a square.

A9. The independent table has all four entries $1/4$, giving $P(X = Y) = 1/2$. A dependent table has entries $p(0,0) = p(1,1) = 1/2$ and the two off-diagonal entries zero, giving $P(X = Y) = 1$. Both have the required marginals. Thus marginals alone do not determine a joint distribution.

A10. The support area is $\int_0^1 (1 - x^2) \, \mathrm{d}x = 2/3$, so $c = 3/2$. For fixed y, $0 < x < \sqrt{y}$, giving $f_Y(y) = \frac{3}{2}\sqrt{y}$ on $(0,1)$. The two setups are

$$\int_0^{1/4} \int_0^{\sqrt{y}} \frac{3}{2} \, \mathrm{d}x \, \mathrm{d}y = \int_0^{1/2} \int_{x^2}^{1/4} \frac{3}{2} \, \mathrm{d}y \, \mathrm{d}x = y^{3/2} \Big|_0^{1/4} = \frac{1}{8}.$$

11.2 B. Expectation, LOTUS, and properties

B1. The four unnormalized masses are $1, 2, 5, 10$, totaling 18. Hence $c = 1/18$ and

$$\mathbb{E}[X] = \frac{0 + 2 + 10 + 30}{18} = \frac{7}{3}, \qquad \mathbb{E}[2X + 3] = 2\left(\frac{7}{3}\right) + 3 = \frac{23}{3}.$$

B2. By LOTUS, $\mathbb{E}[X] = \int_0^1 3x^3 \, \mathrm{d}x = 3/4$ and $\mathbb{E}[1/X] = \int_0^1 3x \, \mathrm{d}x = 3/2$. The latter differs from $1/\mathbb{E}[X] = 4/3$ because expectation is linear, but taking a reciprocal is a nonlinear operation.

B3. The density is even, so the odd integrand $x(1 - |x|)$ integrates to zero. For the other moments,

$$\mathbb{E}[|X|] = 2\int_0^1 x(1-x) \, \mathrm{d}x = \frac{1}{3}, \qquad \mathbb{E}[X^2] = 2\int_0^1 x^2(1-x) \, \mathrm{d}x = \frac{1}{6}.$$

B4. Apply LOTUS to X: $\mathbb{E}[Y] = \mathbb{E}[X^2] = (0 + 2 + 4)/4 = 3/2$ and $\mathbb{E}[Y^2] = \mathbb{E}[X^4] = (0 + 2 + 16)/4 = 9/2$. Thus $\mathrm{Var}(Y) = 9/2 - (3/2)^2 = 9/4$. Squaring $\mathbb{E}[X]$ would not give $\mathbb{E}[X^2]$.

B5. The continuous LOTUS formula is $\mathbb{E}[g(X)] = \int g(x) f_X(x) \, \mathrm{d}x$, when the expectation exists. Here

$$\mathbb{E}[(1 - X)^2] = 2\int_0^1 (x - 2x^2 + x^3) \, \mathrm{d}x = \frac{1}{6}, \qquad \mathbb{E}[\log X] = 2\int_0^1 x \log x \, \mathrm{d}x = -\frac{1}{2}.$$

For the logarithm, integration by parts gives $\int_0^1 x \log x \, \mathrm{d}x = -1/4$; the endpoint limit $x^2 \log x \to 0$ is used.

B6. Profit is $G = 100 - 20X$. Since $\mathbb{E}[X] = 0(1/2) + 1(3/10) + 3(1/5) = 9/10$, linearity gives $\mathbb{E}[G] = 100 - 20(9/10) = 82$ dollars.

B7. Directly, the four values of $3X-2Y+5$ are $5, 1, 8, 4$, so the expectation is $(5+2+24+16)/10 = 47/10$. Alternatively, $\mathbb{E}[X] = 7/10$ and $\mathbb{E}[Y] = 12/10$, giving $3(7/10)-2(12/10)+5 = 47/10$. No independence was needed.

B8. Linearity gives $3(-2) - \frac{1}{2}(4) + 7 = -1$. Independence is unnecessary. The marginal means alone do not determine $\mathbb{E}[XY]$: for example, let a symmetric $R \in \{-1, 1\}$ have equal probabilities and take $X = -2 + R$, with either $Y = 4 + R$ or $Y = 4 - R$. The two products have expectations -7 and -9, respectively.

B9. Normalization is $\int_1^\infty x^{-2}\,\mathrm{d}x = 1$. The mean is the divergent nonnegative integral $\int_1^\infty x^{-1}\,\mathrm{d}x = \infty$. Nevertheless $\mathbb{E}[1/X] = \int_1^\infty x^{-3}\,\mathrm{d}x = 1/2$. A transformed expectation can be finite when the original mean is not.

B10. Since I is an indicator,

$$\mathbb{E}[I] = P(X > 1/2) = \int_{1/2}^1 2x\,\mathrm{d}x = \frac{3}{4}, \qquad \mathbb{E}[XI] = \int_{1/2}^1 2x^2\,\mathrm{d}x = \frac{7}{12}.$$

The second expression is an unnormalized truncated first moment, not a conditional mean.

11.3 C. Raw moments, central moments, and variance

C1. The raw moments are $\mu_1' = (1+4+9)/6 = 7/3$, $\mu_2' = (1+8+27)/6 = 6$, and $\mu_3' = (1+16+81)/6 = 49/3$. Thus

$$\mathrm{Var}(X) = 6 - 49/9 = \frac{5}{9}, \qquad \mu_3 = \mu_3' - 3\mu\mu_2' + 2\mu^3 = \frac{49}{3} - 42 + \frac{686}{27} = -\frac{7}{27}.$$

C2. Integration gives $\mathbb{E}[X^r] = 4\int_0^1 x^{r+3}\,\mathrm{d}x = 4/(r+4)$. Hence $\mu = 4/5$, $\mu_2' = 2/3$, $\mu_3' = 4/7$. The requested central moments are

$$\mathrm{Var}(X) = \frac{2}{3} - \frac{16}{25} = \frac{2}{75}, \qquad \mu_3 = \frac{4}{7} - 3\left(\frac{4}{5}\right)\left(\frac{2}{3}\right) + 2\left(\frac{4}{5}\right)^3 = -\frac{4}{875}.$$

C3. With $\mu = 2$, the variance is $7 - 2^2 = 3$. Expanding the central powers gives

$$\mu_3 = 26 - 3(2)(7) + 2(2^3) = 0,$$

$$\mu_4 = 107 - 4(2)(26) + 6(2^2)(7) - 3(2^4) = 19.$$

These are central moments, whereas the supplied four numbers were raw moments.

C4. First $\mathbb{E}[X^2] = \mathrm{Var}(X) + (\mathbb{E}[X])^2 = 5 + 4 = 9$. Linearity gives $\mathbb{E}[Y] = 3(2) - 4 = 2$, and scaling gives $\mathrm{Var}(Y) = 3^2(5) = 45$. Therefore $\mathbb{E}[Y^2] = 45 + 2^2 = 49$.

C5. The density is $1/4$ on $(-2, 2)$, so

$$\mathbb{E}[Y] = \mathbb{E}[X^2] = \frac{1}{4}\int_{-2}^2 x^2\,\mathrm{d}x = \frac{4}{3}, \qquad \mathbb{E}[Y^2] = \mathbb{E}[X^4] = \frac{1}{4}\int_{-2}^2 x^4\,\mathrm{d}x = \frac{16}{5}.$$

Thus $\mathrm{Var}(Y) = 16/5 - 16/9 = 64/45$.

C6. By linearity, $\mathbb{E}[Z] = (\mathbb{E}[X] - \mu)/\sigma = 0$. Since a shift does not change variance and scaling squares its multiplier, $\mathrm{Var}(Z) = \mathrm{Var}(X)/\sigma^2 = 1$. Rearranging the variance identity gives $\mathbb{E}[X^2] = \sigma^2 + \mu^2$.

C7. The mean exists: $\mathbb{E}[X] = \int_1^\infty 2x^{-2}\,\mathrm{d}x = 2$. But $\mathbb{E}[X^2] = \int_1^\infty 2x^{-1}\,\mathrm{d}x = \infty$. Since the mean is finite, the central second moment also diverges, so $\mathrm{Var}(X) = \infty$. The divergent integral is the second-moment integral.

C8. Symmetry gives $\mu = 0$. Then $\sigma^2 = (4+1+0+1+4)/5 = 2$ and $\mu_4 = (16+1+0+1+16)/5 = 34/5$. The standardized fourth moment is $(34/5)/2^2 = 17/10$.

C9. Write $X - 5 = (X-3) - 2$. Since $\mathbb{E}[X-3] = 0$,

$$\mathbb{E}[(X-5)^2] = \mathbb{E}[(X-3)^2] - 4\mathbb{E}[X-3] + 4 = 4 + 4 = 8.$$

Variance centers at the actual mean 3, whereas this expression centers at 5.

C10. The odd first-moment integrand integrates to zero. For even powers, $\mathbb{E}[X^{2k}] = 3\int_0^1 x^{2k+2}\,\mathrm{d}x = 3/(2k+3)$. Thus $\mathrm{Var}(X) = \mathbb{E}[X^2] = 3/5$, $\mathbb{E}[Y] = 3/5$, and

$$\mathrm{Var}(Y) = \mathbb{E}[X^4] - (\mathbb{E}[X^2])^2 = \frac{3}{7} - \frac{9}{25} = \frac{12}{175}.$$

The second and fourth raw moments of X are needed for $\mathrm{Var}(X^2)$.

11.4 D. Moment generating functions

D1. By definition,

$$M_X(t) = 5\int_0^\infty e^{-(5-t)x}\,\mathrm{d}x = \frac{5}{5-t}, \qquad t < 5.$$

The integral diverges for $t \geq 5$. Its first two derivatives are $5/(5-t)^2$ and $10/(5-t)^3$. At zero they give $\mathbb{E}[X] = 1/5$, $\mathbb{E}[X^2] = 2/25$, and variance $2/25 - 1/25 = 1/25$.

D2. The MGF is $M_X(t) = \frac{1}{4}e^{-t} + \frac{1}{2} + \frac{1}{4}e^{2t}$, finite for every real t. Differentiation gives $M_X'(0) = -1/4 + 1/2 = 1/4$ and $M_X''(0) = 1/4 + 1 = 5/4$. Hence $\mathrm{Var}(X) = 5/4 - 1/16 = 19/16$.

D3. Use $M_{aX+b}(t) = e^{bt}M_X(at)$:

$$M_Z(t) = e^{-2t}M_X(t/3) = e^{-2t}\exp\left(2t + \frac{t^2}{2}\right) = e^{t^2/2}.$$

Thus $M_Z'(0) = 0$ and $M_Z''(0) = 1$, giving mean zero and variance one.

D4. Differentiation yields $M_X'(t) = 6(1-2t)^{-4}$ and $M_X''(t) = 48(1-2t)^{-5}$. Therefore $\mathbb{E}[X] = 6$, $\mathbb{E}[X^2] = 48$, and $\mathrm{Var}(X) = 48 - 36 = 12$. A shift multiplies the MGF by an exponential: $M_{X+4}(t) = e^{4t}(1-2t)^{-3}$ for $t < 1/2$.

D5. From the uniform density,

$$M_X(t) = \frac{1}{2}\int_0^2 e^{tx}\,\mathrm{d}x = \begin{cases} (e^{2t}-1)/(2t), & t \neq 0, \\ 1, & t = 0. \end{cases}$$

It is finite for all real t. Expanding gives $M_X(t) = 1 + t + \frac{2}{3}t^2 + O(t^3)$. Comparing with $1 + \mathbb{E}[X]t + \mathbb{E}[X^2]t^2/2 + \cdots$ gives $\mathbb{E}[X] = 1$ and $\mathbb{E}[X^2] = 4/3$.

D6. Independence permits multiplication:

$$M_S(t) = M_X(2t)M_Y(t) = \frac{2}{2-2t}\frac{4}{4-t} = \frac{4}{(1-t)(4-t)}, \qquad t < 1.$$

Both factors must exist, so $t < 1$ and $t < 4$ are intersected. Direct moment properties give $\mathbb{E}[S] = 2(1/2) + 1/4 = 5/4$ and $\mathrm{Var}(S) = 4(1/4) + 1/16 = 17/16$.

D7. Equality on a neighborhood of zero implies identical distributions. The MGF is that of an exponential variable with rate 2, so $P(X > 1) = \int_1^\infty 2e^{-2x}\,\mathrm{d}x = e^{-2}$. This identifies the marginal law; it gives no independence information about a pair of variables. They could even be equal almost surely.

D8. Transformation gives

$$M_W(t) = e^{3t}M_X(-2t) = \frac{e^{3t}}{1+2t}, \qquad -2t < 1 \iff t > -1/2.$$

The original mean and variance are both 1, so $\mathbb{E}[W] = 3 - 2 = 1$ and $\mathrm{Var}(W) = (-2)^2 = 4$. The domain reverses direction because the scale coefficient is negative.

D9. At zero, $M'(0) = 2$ and $M''(0) = 2$. An MGF would therefore imply $\mathrm{Var}(X) = M''(0) - (M'(0))^2 = 2 - 4 = -2$, which is impossible. The conditions $M(0) = 1$ and positivity of $M(t)$ are not sufficient by themselves.

D10. For fixed $t > 0$, exponential growth eventually dominates x^2, so $e^{tx}/x^2 \geq 1$ for all sufficiently large x. Therefore $\int_1^\infty e^{tx}x^{-2}\,\mathrm{d}x = \infty$. Although $M(0) = 1$, there is no open interval around zero on which it is finite. The usual MGF uniqueness and moment-derivative theorem requires such a neighborhood.

11.5 E. Product moments, covariance, and correlation

E1. The marginal success probabilities give $\mathbb{E}[X] = 1/2$, $\mathbb{E}[Y] = 7/10$, and the only nonzero product contributes $\mathbb{E}[XY] = p(1,1) = 2/5$. Hence $\mathrm{Cov}(X,Y) = 2/5 - 7/20 = 1/20$. The variances are $1/4$ and $21/100$, so

$$\mathrm{Corr}(X,Y) = \frac{1/20}{\sqrt{(1/4)(21/100)}} = \frac{1}{\sqrt{21}}.$$

Independence fails since $p(1,1) = 2/5 \neq (1/2)(7/10) = 7/20$.

E2. The marginals are $f_X(x) = 2x$ and $f_Y(y) = 2y$, both on $(0,1)$. Their product equals $4xy$ throughout the rectangular support, proving independence. Since both means are $2/3$,

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] = \frac{4}{9}, \qquad \mathrm{Cov}(X,Y) = 0.$$

Each variance is $1/2 - 4/9 = 1/18 > 0$, so correlation is also zero.

E3. The marginals from A3 give $\mathbb{E}[X] = \int_0^1 2x^2\,\mathrm{d}x = 2/3$ and $\mathbb{E}[Y] = \int_0^1 2y(1-y)\,\mathrm{d}y = 1/3$. Also

$$\mathbb{E}[XY] = \int_0^1 \int_0^x 2xy\,\mathrm{d}y\,\mathrm{d}x = \int_0^1 x^3\,\mathrm{d}x = \frac{1}{4}.$$

Thus covariance is $1/4 - 2/9 = 1/36$. The second moments are $1/2$ and $1/6$, so both variances equal $1/18$; correlation is $(1/36)/(1/18) = 1/2$.

E4. Symmetry gives $\mathbb{E}[X] = \mathbb{E}[X^3] = 0$, while $\mathbb{E}[Y] = \mathbb{E}[X^2] = 1/3$. Therefore $\operatorname{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$. But $A = B$ as events, with probability $1/2$, so $P(A \cap B) = 1/2 \neq (1/2)^2$. Thus X and Y are dependent despite zero covariance.

E5. The covariance identity gives $\operatorname{Cov}(X, Y) = -5 - (2)(-1) = -3$. Dividing by the product of standard deviations gives $\operatorname{Corr}(X, Y) = -3/(2 \cdot 3) = -1/2$.

E6. Covariance is linear in its second argument: $\operatorname{Cov}(X, 5 - 3X) = -3 \operatorname{Var}(X) = -6$. Also $\operatorname{Var}(Y) = 9 \operatorname{Var}(X) = 18$. Hence $\operatorname{Corr}(X, Y) = -6/\sqrt{2 \cdot 18} = -1$. The negative coefficient describes an exact decreasing affine relationship.

E7. Constants have zero covariance with every variable, so

$$\operatorname{Cov}(2X + 7, -4Y + 1) = (2)(-4) \operatorname{Cov}(X, Y) = (-8)(-2) = 16.$$

Shifts disappear from covariance; the two scale coefficients remain as a product.

E8. Since $X^2 = X$, $\mathbb{E}[X^2] = \mathbb{E}[X] = 2/5$. For Y, $\mathbb{E}[Y] = 1/2$ and $\mathbb{E}[Y^2] = 2/2^2 = 1/2$. Independence allows factorization of functions of the two variables:

$$\mathbb{E}[X^2 Y] = \frac{2}{5}\frac{1}{2} = \frac{1}{5}, \qquad \mathbb{E}[XY^2] = \frac{2}{5}\frac{1}{2} = \frac{1}{5}.$$

E9. Cauchy–Schwarz gives $|\operatorname{Cov}(X, Y)| \leq \sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)} = 2$. The specified value 3 violates this bound and would give correlation $3/2$. The allowable interval is $[-2, 2]$.

E10. For nonzero a, c and positive variances,

$$\operatorname{Corr}(aX + b, cY + d) = \frac{ac \operatorname{Cov}(X, Y)}{|a|\sigma_X |c|\sigma_Y} = \operatorname{sgn}(ac) \operatorname{Corr}(X, Y).$$

Both scales are negative in the first pair, so the result is $-2/5$. Their signs differ in the second pair, so the result is $2/5$.

11.6 F. Moments of one and two linear combinations

F1. Linearity gives $\mathbb{E}[U] = 2(3) - 3(-2) + 5 = 17$. Variance includes the covariance term:

$$\operatorname{Var}(U) = 2^2(4) + (-3)^2(9) + 2(2)(-3)(1) = 16 + 81 - 12 = 85.$$

The added constant affects the mean but not the variance.

F2. Combine the payouts first: $W + V = 1.5X + 2Y + 30$. Therefore

$$\operatorname{Var}(W + V) = (1.5)^2(100) + 2^2(64) + 2(1.5)(2)(20) = 225 + 256 + 120 = 601.$$

F3. With coefficients $(2, -1, 3)$, the variance is

$$\begin{aligned} \operatorname{Var}(2X - Y + 3Z) &= 4(1) + 1(4) + 9(9) \\ &\quad + 2(2)(-1)(1/2) + 2(2)(3)(-1) + 2(-1)(3)(1) \\ &= 4 + 4 + 81 - 2 - 12 - 6 = 69. \end{aligned}$$

Each unordered pair appears once, multiplied by two and by its signed coefficients.

F4. The variances are $\mathrm{Var}(U) = 2 + 4(5) + 4(-1) = 18$ and $\mathrm{Var}(V) = 9(2) + 5 - 6(-1) = 29$. Bilinearity gives

$$\mathrm{Cov}(U, V) = 3\,\mathrm{Var}(X) - 2\,\mathrm{Var}(Y) + (6-1)\,\mathrm{Cov}(X, Y) = 6 - 10 - 5 = -9.$$

Therefore $\mathrm{Corr}(U, V) = -9/\sqrt{18 \cdot 29} = -9/\sqrt{522}$.

F5. For $S = \sum_{i=1}^{5} X_i$, linearity gives $\mathbb{E}[S] = 20$, and the covariance terms vanish, giving $\mathrm{Var}(S) = 15$. For $\overline{X} = S/5$, the mean is 4 and variance is $15/25 = 3/5$. All four answers remain valid under pairwise zero covariance; even the zero-covariance assumption is unnecessary for the two means.

F6. The moments of X are $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 1/3$, $\mathbb{E}[X^3] = 0$, and $\mathbb{E}[X^4] = 1/5$. Thus $\mathrm{Var}(Y) = 1/5 - 1/9 = 4/45$ and $\mathrm{Cov}(X, Y) = 0$. Consequently

$$\mathrm{Var}(2X + Y) = 4\left(\frac{1}{3}\right) + \frac{4}{45} + 4(0) = \frac{64}{45}.$$

The covariance vanishes by symmetry, while $Y = X^2$ still determines dependence.

F7. The sum identity gives $20 = 4 + 9 + 2\,\mathrm{Cov}(X, Y)$, so $\mathrm{Cov}(X, Y) = 7/2$. Then $\mathrm{Var}(X - Y) = 4 + 9 - 2(7/2) = 6$. The covariance contribution changes sign for a difference.

F8. Expand and complete the square:

$$\mathrm{Var}(X - aY) = 9 + 4a^2 - 4a = 8 + 4(a - 1/2)^2.$$

The minimum is 8, attained at $a = 1/2$. This is a variance calculation; no independence assumption is used.

F9. The covariance is $(1/2)\sqrt{25}\sqrt{16} = 10$. The mean is $0.4(100) + 0.6(80) = 88$, and

$$\mathrm{Var}(T) = 0.4^2(25) + 0.6^2(16) + 2(0.4)(0.6)(10) = 4 + 5.76 + 4.8 = \frac{364}{25}.$$

The decimal value is 14.56.

F10. The variances are $\mathrm{Var}(U) = 1 + 4 + 2(1/2) = 6$ and $\mathrm{Var}(V) = 4 + 9 - 2(1) = 11$. Expanding the two arguments gives

$$\mathrm{Cov}(X + Y, Y - Z) = \mathrm{Cov}(X, Y) - \mathrm{Cov}(X, Z) + \mathrm{Var}(Y) - \mathrm{Cov}(Y, Z) = \frac{1}{2} + 1 + 4 - 1 = \frac{9}{2}.$$

Thus correlation is $(9/2)/\sqrt{66} = 9/(2\sqrt{66})$.

11.7 G. Conditional expectation and conditional variance

G1. At $X = 1$, the unnormalized weights are $1, 2, 3$, totaling 6. Thus $p_{Y|X}(y \mid 1) = (1+y)/6$ for $y = 0, 1, 2$. The conditional first and second moments are

$$\mathbb{E}[Y \mid X = 1] = \frac{0 + 2 + 6}{6} = \frac{4}{3}, \qquad \mathbb{E}[Y^2 \mid X = 1] = \frac{0 + 2 + 12}{6} = \frac{7}{3}.$$

Their difference gives conditional variance $7/3 - 16/9 = 5/9$.

G2. Integrating over x gives $f_Y(y) = \frac{2}{3}(1+y)$ on $(0,1)$, and hence $f_{X|Y}(x \mid y) = (2x+y)/(1+y)$ on $(0,1)$. At $y = 1/2$,

$$\mathbb{E}[X \mid Y = 1/2] = \frac{\int_0^1 x(2x+1/2)\ \mathrm{d}x}{3/2} = \frac{2/3+1/4}{3/2} = \frac{11}{18},$$

$$\mathbb{E}[X^2 \mid Y = 1/2] = \frac{1/2+1/6}{3/2} = \frac{4}{9}.$$

Thus conditional variance is $4/9 - (11/18)^2 = 23/324$.

G3. The marginal $f_X(x) = 2x$ gives $f_{Y|X}(y \mid x) = 1/x$ on $(0,x)$. This is uniform, so $\mathbb{E}[Y \mid X = x] = x/2$ and $\mathrm{Var}(Y \mid X = x) = x^2/12$. At $x = 4/5$, the answers are $2/5$ and $(16/25)/12 = 4/75$.

G4. Independence makes the conditional density equal to f_X. Therefore $\mathbb{E}[X \mid Y = y] = \int_0^1 2x^2\ \mathrm{d}x = 2/3$, while the conditional second moment is $\int_0^1 2x^3\ \mathrm{d}x = 1/2$. The conditional variance is $1/2 - 4/9 = 1/18$, independent of y.

G5. The Bernoulli conditional law gives $\mathbb{E}[Y \mid X] = X$ and $\mathrm{Var}(Y \mid X) = X(1-X)$. Total expectation gives $\mathbb{E}[Y] = \mathbb{E}[X] = 1/2$. Total variance gives

$$\mathrm{Var}(Y) = \mathbb{E}[X(1-X)] + \mathrm{Var}(X) = \left(\frac{1}{2} - \frac{1}{3}\right) + \frac{1}{12} = \frac{1}{4}.$$

Both the variation within conditional groups and variation between their means contribute.

G6. At fixed y, the conditional density is $1/(1-y)$ for $y < x < 1$. Hence

$$\mathbb{E}[X \mid Y = y] = \frac{\int_y^1 x\ \mathrm{d}x}{1-y} = \frac{1-y^2}{2(1-y)} = \frac{1+y}{2}.$$

Integrating this conditional mean against the marginal yields $\mathbb{E}[X] = \int_0^1 \frac{1+y}{2}\, 2(1-y)\ \mathrm{d}y = \int_0^1 (1-y^2)\ \mathrm{d}y = 2/3$.

G7. The conditional mean is $Y/2$ and conditional variance is $Y^2/12$. Since Y is uniform on $(0,2)$, $\mathbb{E}[Y] = 1$, $\mathrm{Var}(Y) = 1/3$, and $\mathbb{E}[Y^2] = 4/3$. Therefore

$$\mathbb{E}[X] = \frac{1}{2}, \qquad \mathrm{Var}(X) = \frac{\mathbb{E}[Y^2]}{12} + \mathrm{Var}(Y/2) = \frac{1}{9} + \frac{1}{12} = \frac{7}{36}.$$

G8. Normalization gives $1 = c\int_0^1 x^2\ \mathrm{d}x = c/3$, so $c = 3$. The marginal is $f_X(x) = 3x^2$, hence $f_{Y|X}(y \mid x) = 1/x^2$ for $0 < y < x^2$. At $x = 1/2$, this is uniform on $(0, 1/4)$, and

$$P(Y > 1/8 \mid X = 1/2) = \int_{1/8}^{1/4} 4\ \mathrm{d}y = \frac{1}{2}.$$

G9. The event probability is $P(X > 1/2) = \int_{1/2}^1 2x\ \mathrm{d}x = 3/4$. Using a restricted joint first moment,

$$\mathbb{E}[Y \mid X > 1/2] = \frac{\int_{1/2}^1 \int_0^x 2y\ \mathrm{d}y\ \mathrm{d}x}{3/4} = \frac{\int_{1/2}^1 x^2\ \mathrm{d}x}{3/4} = \frac{7}{18}.$$

In contrast, $\mathbb{E}[Y \mid X = 1/2] = (1/2)/2 = 1/4$. The first averages over every $X > 1/2$; the second uses the conditional density at a single value.

G10. Here $\mathbb{E}[Y] = 3/2$ and $\mathrm{Var}(Y) = 1/4$. Thus $\mathbb{E}[X] = \mathbb{E}[1 + 2Y] = 4$. Total variance gives

$$\mathrm{Var}(X) = \mathbb{E}[3Y] + \mathrm{Var}(1 + 2Y) = \frac{9}{2} + 4\left(\frac{1}{4}\right) = \frac{11}{2}.$$

Finally $\mathbb{E}[X^2] = \mathrm{Var}(X) + (\mathbb{E}[X])^2 = 11/2 + 16 = 43/2$.

12 TI-84 Plus CE: a calculator-first workflow

Use the calculator for the arithmetic, finite sums, numeric integration, and moment calculations wherever the model is known. The accompanying `STAM2` program suite adds guided solvers and an on-calculator encyclopedia. It uses the structure of your earlier `STA431` suite with separate program names. The recipes below are also available through native calculator commands, so you can still work directly from the home screen.

The calculator can produce most numerical answers in Homeworks 4–5 after you identify the support, event, or conditional slice. A symbolic density formula, a convergence domain, or an independence proof still needs the mathematics in the main guide. In particular, finite numerical integration bounds do not stand for infinity, and matching numerical values at a few points does not prove two functions equal.

12.1 Start with the Midterm II suite

Transfer **all ten** `.8xp` files from the package's `calculator_files` directory using TI Connect CE. Then press `PRGM`, select `STAM2`, and press `ENTER` twice. This is a TI-BASIC program suite, launched from `PRGM`; it is not a Flash application in the `APPS` menu.

Main menu	Use it for
DISCRETE + JOINT	A finite PMF or joint table, entered as rows or a formula/grid: normalization, moments, marginals, conditionals, events, and functions.
DENSITY + INTEGRAL	A one-dimensional density or fixed conditional slice; numerical double integrals with support limits you enter.
MGF TOOLS	Exponential and quadratic-exponential templates, affine transformations, and numerical derivatives of a typed MGF.
LINEAR COMBOS	One or two combinations of up to three variables from their given variances and covariances, with means optional.
ENCYCLOPEDIA	Definitions, formula selection, support rules, derivation outlines, and interpretation.
HOMEWORK MAP	Read a problem-by-problem walkthrough, or choose LOAD + SOLVE to prefill the homework model and run its general solver.

Fast table entry. Choose `DISCRETE + JOINT > FORMULA/GRID`. For HW4 Q1, enter x values $\{1, 2\}$, y values $\{1, 2, 3\}$, and $q(X, Y) = X + Y^2$. The program generates all six pairs, sums the weights, and gives $c = 1/37$. For HW5 Q8, use $\{1, 2, 3\}$ in both lists and $q = X + Y$. For a one-variable formula use the y list $\{0\}$; for example x list $\{1, 2, 3, 4\}$ with $q = X^2 + X$ generates HW5 Q1. The grid is limited to 30 pairs; manual row entry also supports finite nonrectangular supports.

Shortest homework route. Choose `HOMEWORK MAP > LOAD + SOLVE`, the homework group, and its question. The program supplies the input model to the existing solver. A double-integral preset asks for an even N; enter 4 for these polynomial homework examples. For changed exam data, use the manual general solver.

Use `BACK` to return through the menus. In reference pages, `ENTER` advances, left arrow returns to the previous page, and `CLEAR` returns to the topic menu. The solvers use scalar variables, strings, and custom lists beginning `M2`; they leave your graph equations and standard lists `L1`–`L6` unchanged. Native recipes below use those standard lists intentionally. The package includes simulator checks and device test inputs; run the device checks on your actual CE before relying on the suite during an exam.

12.2 Key entry that makes every recipe faster

`[2nd][0]` opens `CATALOG`; paste the actual command token. `[2nd][STAT]` opens `LIST`, with `seq(` under `OPS` and `sum(` under `MATH`. `STAT > CALC` contains `1-Var Stats` and `2-Var Stats`. Choose commands by name because menu positions can differ across OS versions.

The comma-separated calculus commands here use `MODE: CLASSIC`. MathPrint shows templates for the same operations. The printed arrow `->` below means the calculator's `STO▶` token. The text `L1` means the actual L_1 token, entered with `[2nd][1]`; similarly for the other lists. Comparisons such as `<=` mean the single $\leq$ token under `[2nd][MATH] > TEST`. Use `(-)` for a negative number and the subtraction key between two quantities.

Store reusable values. A result followed by `->M` remains in M until overwritten; `Ans` changes after another calculation. Keep a mean in M, a total weight in S or D, and reuse them. In a native recipe, do not change lists or run a program in the middle of a calculation and assume these variables still hold their previous values.

12.3 The fastest finite-PMF trick: unnormalized weighted Stats

Enter possible values in L_1 and their **nonnegative weights** in L_2, with at least one positive weight. They need not sum to one. Run

```
1-Var Stats L1,L2
```

In the wizard set `List=L1` and `FreqList=L2`. Read

$$\bar{x} = \mathbb{E}X, \qquad \sigma_x^2 = \operatorname{Var} X, \qquad n = \sum_i w_i.$$

The population standard deviation is σ_x. The sample S_x is a different quantity; it can be undefined when frequency entries are fractional. Squaring σ_x gives the variance of the modeled distribution. If $p_i = cw_i$, the normalizing constant is $c = 1/n$.

Homework 5, Problem 1, with almost no arithmetic

Enter $L_1 = \{1, 2, 3, 4\}$ and $L_2 = \{2, 6, 12, 20\}$. Do not divide the second list by 40. `1-Var Stats L1,L2` gives $n = 40$, $\bar{x} = 13/4$, $\sigma_x^2 = 63/80$. Therefore $c = 1/40$, $\mathbb{E}X = 13/4$, and $\operatorname{Var} X = 63/80$.

Transform the values and reuse the weights. To obtain the mean and variance of $Y = X^2$ from a discrete distribution:

```
L1^2->L3
1-Var Stats L3,L2
```

Now $\bar{x}$ on the results screen is $\mathbb{E}Y$, and σ_x^2 is $\operatorname{Var} Y$. More generally, put $g(L_1)$ into L_3 and use the same weight list. Do not square the original variance to obtain the variance of a square.

For raw and central moments directly from the home screen, choose an integer power R:

```
sum(L1^R*L2)/sum(L2)
sum((L1-M)^R*L2)/sum(L2)
```

The second expression assumes that the original weighted mean was stored in M. This computes third and fourth central moments directly, avoiding a long raw-to-central expansion. Keep the expansion in your written work if the question explicitly requests that derivation.

12.4 A universal joint-table setup

Use **one row for every allowed pair**: L_1 contains x, L_2 the matching y, and L_3 the nonnegative raw weight. Repeated coordinates are expected. The pair list need not be a rectangular grid; list exactly the support. Set

```
sum(L3)->S
```

Then $c = 1/S$ when the input is an unnormalized weight. For product moments,

```
sum(L1^R*L2^T*L3)/S
```

returns $\mathbb{E}[X^R Y^T]$. Setting $R = T = 1$ gives $\mathbb{E}XY$. A table-based linear combination can be handled by transforming the row values, without manually expanding its covariance terms:

```
2*L1-3*L2+5->L4
1-Var Stats L4,L3
```

Read its mean and population variance.

Homework 5, Problem 8, entered once

```
{1,1,1,2,2,2,3,3,3}->L1
{1,2,3,1,2,3,1,2,3}->L2
L1+L2->L3
sum(L3)->S
```

Here $S = 36$. `1-Var Stats L1,L3` gives $\mathbb{E}X = 13/6$ and $\operatorname{Var} X = 23/36$; exchanging L_1, L_2 gives the same Y moments. The raw product command `sum(L1*L2*L3)/S` gives $14/3$.

12.5 Masks turn events and conditionals into list calculations

A list comparison returns 1 for a true row and 0 for a false row. Multiply the mask by the weight list to keep exactly the desired rows:

```
sum((L1>L2)*L3)/S
sum((L1+L2<=3)*L3)/S
sum((L1<=A)*(L2<=B)*L3)/S
```

These compute $P(X > Y)$, $P(X + Y \le 3)$, and the joint CDF $F(A, B)$. Compound masks multiply the parenthesized conditions, so their meaning is intersection. To find $P(X = x_0)$, store x_0 in A and use `sum((L1=A)*L3)/S`.

For a conditional distribution, turn the selected joint weights into a new frequency list. For $X \mid Y = A$:

```
L3*(L2=A)->L4
1-Var Stats L1,L4
```

The weights automatically renormalize inside Stats. Read the conditional mean and population variance. Require `sum(L4)>0`. For $Y \mid X = A$, mask with `L1=A` and analyze L_2 instead. A condition such as $X > 1$ can replace equality, which also handles conditioning on interval events.

Finish Homework 5, Problem 8

With the nine-row data above, set $A = 1$, calculate `L3*(L1=A)->L4`, then run `1-Var Stats L2,L4`. The conditional mean is $20/9$, and the conditional variance is $50/81$. The selected weight total is 9, so $P(X = 1) = 9/36 = 1/4$ and $p_{Y|X}(y \mid 1) = (1 + y)/9$ on $y = 1, 2, 3$.

12.6 Two-variable Stats gives covariance with little typing

For the same joint row lists, run

```
2-Var Stats L1,L2,L3
```

Scroll to the means, population standard deviations, total frequency n, and weighted sum Σxy. Then

$$\mathbb{E}XY = \frac{\Sigma xy}{n}, \qquad \mathrm{Cov}(X, Y) = \frac{\Sigma xy}{n} - \bar{x}\bar{y}, \qquad \rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_x \sigma_y}.$$

Retrieve actual output tokens through `VARS > Statistics`, under `XY` or the sum menu. Store values before editing the lists or starting another analysis, which clears the statistics variables. The n here is the *sum of weights*, not necessarily the number of rows. No regression or `DiagnosticsOn` is needed.

For HW5 Problem 8 this yields covariance $-1/36$ and correlation $-1/23$. A direct centered alternative uses only ordinary lists:

```
sum(L1*L3)/S->M
sum(L2*L3)/S->N
sum((L1-M)*(L2-N)*L3)/S
```

This centered calculation often avoids subtracting two similar large values.

12.7 One reusable continuous-density engine

In `Y=`, save a **nonnegative unnormalized shape** $q(X)$ in Y_1. Store finite lower and upper support limits in B, C. For a conditional slice, replace the given variable by a scalar parameter A and hold it fixed. Paste the actual Y_1 token from `VARS > Y-VARS > Function`.

```
fnInt(Y1,X,B,C)->D
fnInt(X*Y1,X,B,C)/D->M
fnInt((X-M)^2*Y1,X,B,C)/D
```

The results are the total weight D, the mean M, and the variance. The normalizing constant is $1/D$ if this is the entire unnormalized density. For a slice of a globally normalized joint PDF, D is instead the appropriate marginal density at the given value. Dividing by D produces the conditional moments automatically. Require $D > 0$.

For any payoff $g(X)$, compute `fnInt(g(X)*Y1,X,B,C)/D`. For its variance, first store that mean in M, then use `fnInt((g(X)-M)^2*Y1,X,B,C)/D`. Here `g(X)` is a placeholder: type the actual expression or a stored function token.

Homework 5, Problem 7, without deriving the denominator numerically

Set $Y_1 = X + A$, $A = 1/2$, $B = 0$, $C = 1$. The engine returns $D = 1$, mean $7/12$, and variance $11/144$. For the *symbolic* conditional density write $(x+y)/(1/2+y)$ on $(0,1)$; the calculator supplies the numerical moments at the chosen y.

For HW5 Problem 2, you may omit the constant $3/4$: use $Y_1 = 1 - X^2$ on $(-1,1)$ and let the ratios normalize it. The second raw moment is $1/5$. To get $\mathrm{Var}(X^2)$ directly:

```
fnInt(X^2*Y1,X,B,C)/D->M
fnInt((X^2-M)^2*Y1,X,B,C)/D
```

This gives $8/175$. The raw fourth moment is `fnInt(X^4*Y1,X,B,C)/D` $= 3/35$.

For an interval $[E,F]$ inside the support, $P(E \le X \le F) =$ `fnInt(Y1,X,E,F)/D`. Clip bounds to the support first. If the density has different formulas on different intervals, split the integral and add the parts. Each new parameter value A requires recomputing D and any stored mean.

12.8 Joint slices: numerical marginals and conditional laws

Choose `DENSITY + INTEGRAL > JOINT SLICE`, then the conditioning direction. Enter the joint shape $q(X,Y)$, its known multiplier c, the given value, and the support bounds for the remaining variable. Use $c = 1$ if q is already the joint PDF. The output includes the appropriate marginal-density value and the conditional mean, second moment, and variance. Further options evaluate a conditional density at a point, integrate a conditional interval, or change the given value without retyping the joint model. This directly supports HW4 Q2, HW5 Q7, and the optional reimbursement question. Symbolic density formulas still follow from the integrals in the walkthrough.

12.9 Double integrals: let the app handle the second numerical layer

The CE cannot nest `fnInt`. The app's `DOUBLE INTEGRAL` tool therefore performs one numerical integral in Y at each outer X node and combines those values with Simpson's rule. Enter:

1. The density or weight $q(X,Y)$ and fixed parameter A (enter 0 if absent).

2. Constant outer x limits, followed by inner y limits as expressions in X.

3. A multiplier $g(X,Y)$: enter 1 for probability or normalization, X, Y, or XY for moments.

4. An even outer resolution N; compare the N and $2N$ results shown.

For an integral $I = \int_a^b G(x)\,\mathrm{d}x$, $G(x) = \int_{L(x)}^{U(x)} q(x,y)\,\mathrm{d}y$, the outer rule is

$$I_N \approx \frac{h}{3}\left[G(x_0) + G(x_N) + 4\sum_{i\ \text{odd}} G(x_i) + 2\sum_{\substack{i\ \text{even}\\ 0<i<N}} G(x_i)\right], \quad h = \frac{b-a}{N}.$$

The app reports **raw integrals**. If q is a globally normalized PDF and the entered bounds describe an event, its raw integral is the event probability. Do not renormalize that event to one. Use the optional normalization ratio only when the entered region is the full support of the model you intend to normalize.

Homework task	Input region and integrand	Result
HW4 Q2 normalizer	$q = 2X + Y$, $x : 0 \to 1$, $y : 0 \to 2$, $g = 1$	Raw $I = 4$, $c = 1/4$
HW4 Q3 survival	$q = (X + Y)/8$, $x : 1 \to 2$, $y : 1 \to 2$, $g = 1$	$I = 3/8$, failure $1 - I = 5/8$
HW4 Q5 event	$q = (2X + Y)/4$, $x : 0 \to 1$, $y : X \to 2$, $g = 1$	$I = 19/24$
Triangle example	$q = 2$, $x : 0 \to 1$, $y : 0 \to X$, $g = XY$	$\mathbb{E}XY = 1/4$

These polynomial cases are particularly favorable. More difficult integrands can be slow or sensitive to bounds; split changing boundaries and compare refinements. Agreement at two resolutions is useful numerical evidence, not an error guarantee or a proof of normalization. If a formula is undefined at an endpoint, use a valid limiting form or a suitable change of variables rather than entering an undefined value.

The native fallback is to do one integral algebraically and call `fnInt` on the remaining expression. For HW4 Q5 it reduces to

```
fnInt(1/2+X-5*X^2/8,X,0,1)
```

which gives 19/24. For a marginal density at a *fixed* conditioning value, a single `fnInt` usually suffices directly.

12.10 Covariance matrices automate all cross terms

The guided `LINEAR COMBOS` solver asks for up to three variances and the distinct pairwise covariances, then coefficient lists for one or two combinations. If means are not supplied, choose the variance-only option; no invented means are needed.

The native matrix equivalent is also fast. Open `[2nd][`x^{-1}`] MATRIX > EDIT`. Put the covariance matrix in $[A]$ and the coefficient rows in $[B]$. Then evaluate

$$[B] * [A] * [B]^{\mathsf{T}}.$$

Paste actual matrix names from `MATRIX > NAMES`; obtain the postfix transpose ^{T} from `MATRIX > MATH`, not the letter `T`. The diagonal outputs are variances and the off-diagonal outputs are covariances. Additive offsets do not belong in the covariance matrix or coefficient rows.

For Homework 5 Problem 6,

$$[A] = \begin{pmatrix} 4 & 1 & -1 \\ 1 & 9 & 0 \\ -1 & 0 & 2 \end{pmatrix}, \qquad [B] = \begin{pmatrix} 3 & -2 & 1 \end{pmatrix},$$

and the result is [[56]]. For two combinations $U = 2X - Y + 1$, $V = X + 3Y - 4$, with $\operatorname{Var} X = 4$, $\operatorname{Var} Y = 9$, $\operatorname{Cov}(X, Y) = -1$,

$$[A] = \begin{pmatrix} 4 & -1 \\ -1 & 9 \end{pmatrix}, \quad [B] = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix} \quad \Longrightarrow \quad \begin{pmatrix} 29 & -24 \\ -24 & 79 \end{pmatrix}.$$

This gives $\rho_{UV} = -24/\sqrt{29 \cdot 79}$. In written work, retain the general variance or covariance formula so the meaning of the entries is visible.

12.11 MGFs: exact templates first, numerical derivatives when needed

The `MGF TOOLS` exponential template takes rate λ and supplies $M(t) = \lambda/(\lambda - t)$, $t < \lambda$, together with mean $1/\lambda$, second moment $2/\lambda^2$, and variance $1/\lambda^2$. The quadratic-exponential template $e^{\alpha t+\beta t^2}$ gives mean α and variance 2β. These directly cover HW5 Problems 3–4. The affine transformation uses the scale a and shift b in $Z = aX + b$.

For a generic given MGF, save it in Y_1, with calculator X representing mathematical t. The mean can be estimated by `nDeriv(Y1,X,0)`. A numerical second moment can be estimated by centered differences:

$$\mathbb{E}X \approx \frac{M(h) - M(-h)}{2h}, \qquad \mathbb{E}X^2 \approx \frac{M(h) - 2M(0) + M(-h)}{h^2}.$$

The generic app compares two small step sizes and labels these estimates. Keep stable digits; a small step can worsen cancellation, so smaller is not automatically better. The points used must lie inside the MGF's domain. The exact template is preferable when it matches the expression.

For a native affine transform, put a in A, b in B, and define Y_2 by

```
e^(B*X)*Y1(A*X)
```

This evaluates $M_{aX+b}(t)$ numerically at any valid t. It does not symbolically simplify the expression. With $M_X(t) = e^{4t+8t^2}$, $a = 1/4$, $b = -1$, simplify on paper to $e^{t^2/2}$; the app's exact template then gives mean 0 and variance 1.

For a finite-support PDF in Y_1, the native expression

```
fnInt(e^(T*X)*Y1,X,B,C)/D
```

gives the MGF value at a chosen numeric T. This is particularly useful for checking a proposed MGF against the original distribution.

12.12 A useful extra: minimize a linear-combination variance

Practice question F8 asks for the coefficient a minimizing $\mathrm{Var}(X - aY)$ when the two variances are 9 and 4 and covariance is 2. The quadratic is $9 + 4a^2 - 4a$. The native command

```
fMin(9+4X^2-4X,X,-5,5)
```

returns the minimizing coefficient $X = 1/2$, not the minimum variance itself. Substitute it into the quadratic to obtain 8. Choose a search interval containing the relevant minimum; `fMin` is numerical. For the general exact answer, differentiate to get $a = \mathrm{Cov}(X, Y)/\mathrm{Var}(Y)$ when $\mathrm{Var}(Y) > 0$.

12.13 Official references and validation limits

The commands were checked against the TI-84 Plus CE Reference Guide, including `fnInt`, `nDeriv`, `seq`, `sum`, `fMin`, and one- and two-variable Stats. TI's detailed family guide explains fractional frequency weights and stored statistics. See official support articles 34416 (weighted Stats), 34431 (matrix transpose), 34417 (stored functions), and 34694 (numerical integration). TI's hosted McIntyre, Mathematics on a TI-84/CE, Volume 2, Chapter 15, discusses the non-nesting integral limit and outer quadrature. Consulted October 27, 2026.

The new suite is checked by tokenization, static layout/control-flow checks, and a TI-BASIC simulator against homework targets and edge cases. This verifies those source-level behaviors within the simulator; it does not establish that every physical-CE behavior or arbitrary user-entered integrand has been tested. The transfer package supplies concrete device checks with expected results.

13 Improvements made while walking through the homework

The walkthrough was also a usability test. The following changes were implemented in the calculator package, rather than left as suggestions.

Friction found	Change made	Concrete benefit
Repeated model entry	`LOAD + SOLVE` provides input presets for all sixteen questions, using the same general calculators.	HW5 Q6 needs eleven numeric entries manually and zero through quick load; both produce variance 56.
Individual marginal queries	`MARGINAL TABLES` aggregates and prints both complete finite marginal distributions.	HW4 Q1's five marginal values appear in one command instead of five separate queries.
An unhelpful independence-failure message	The app now displays a violating pair, its joint probability, and its marginal product.	The displayed counterexample gives a concrete starting point for the exact written non-independence proof.
No direct continuous marginal/conditional workflow	The new `JOINT SLICE` tool keeps the original variable names, accounts for a known global multiplier, and reports marginal values and conditional results.	HW4 Q2 and the reimbursement problem can be checked without manually rewriting every conditional density as a one-variable model.
Repeated joint-model entry at a new conditioning value	`NEW GIVEN VALUE` reuses the shape and multiplier, then recomputes the slice with the bounds you enter.	HW5 Q7 can be checked at $y = 1/2$ and another y without retyping $X + Y$.
Dummy-variable output on a one-variable grid	A single y coordinate equal to zero is recognized as a univariate model.	HW5 Q1 no longer shows irrelevant Y moments and undefined-correlation pages.
A stored-function tokenization defect in an optional preset	The triangular density preset stores $1 - \sqrt{X^2}$ using executable calculator tokens.	The optional zero-covariance question loads correctly; this equals $1 - \lvert X \rvert$ for real X.

Where the app saves the most work

Finite-table arithmetic is highly automatable: normalization, all marginal masses, conditional probabilities, first/second moments, covariance, and transformed moments come from one loaded model. Given-moment questions such as HW5 Q5–Q6 are also ideal: the app expands the covariance terms and preserves their signs. The exact MGF templates handle the two assigned MGF models with minimal entry.

For joint continuous problems, the app removes most numerical integration and conditional-normalization arithmetic after the support and region have been selected. It still needs you to identify whether the region is the full support, an event, or a conditional slice. The homework's symbolic marginal formulas and independence proofs are short enough to write from the displayed setups; a numerical value cannot replace those functions or logical arguments.

Useful next improvements

Useful next additions would be a probability/complement shortcut for the device question and an optional support sketch. The current tools provide numerical results and analytical templates; symbolic density formulas and proofs remain in the written solution.

14 Textbook reading map and further practice

Page numbers below are **printed page numbers**, unless explicitly called PDF pages. These references are keyed to the actual supplied files, rather than assumed editions. Explanations and the 70-question bank in this guide are newly written; exercise references point you back to your own textbooks.

14.1 Primary text: Freund, eighth edition

Irwin Miller and Marylees Miller, *John E. Freund's Mathematical Statistics with Applications*, eighth edition, Pearson Education Limited, 2014, supplied custom-library PDF. Its printed page 82 is PDF page 87; in the relevant chapters, add five to a printed page to find the PDF page.

Topic	Read	Focus while reading
Joint laws	Ch3 §5, pp.82–91	Normalization, events in the plane, full CDFs.
Marginals	Ch3 §6, pp.92–95	Sum or integrate out the other variable.
Conditionals; independence	Ch3 §7, pp.95–101	Divide by the conditioning marginal; compare with a product of marginals.
Expectation; LOTUS	Ch4 §2, pp.113–121	Expectations of functions and linearity.
Moments; variance	Ch4 §3, pp.121–125	Raw versus central moments; affine variance.
MGFs	Ch4 §5, pp.127–131	Definition, derivatives, scale/shift rules.
Product moments	Ch4 §6, pp.131–135	$\mathbb{E}XY$, covariance, independence versus uncorrelatedness.
Linear combinations	Ch4 §7, pp.136–138	Full variance expansion and covariance of two combinations.
Conditional moments	Ch4 §8, pp.138–140	Conditional mean and variance; p.140 has exercises.

Freund Ch4 §4 (Chebyshev's theorem) lies between moments and MGFs but is outside your supplied topic list. It is not required to complete this guide. Correlation appears later in Freund Ch6 §7, p.196, in a bivariate-normal discussion; the general treatment here is supported by Bertsekas–Tsitsiklis §4.5 and Casella–Berger §4.5.

Freund practice to pair with this guide. The multivariate section includes a source-checked Chapter 3 exercise list. For Chapter 4, try these selected exercises after the corresponding practice set:

- **Expectation:** Exercises 9 and 10, p.121. Compute raw moments and expectations of polynomial functions.
- **Moments:** Exercises 19, 21, 23–25, p.130. Practice continuous moments, affine variance, standardization, moment existence, and raw-to-central expansions.
- **MGFs:** Exercises 33, 35–40, p.131. Uniform MGF, log-MGF derivatives, invalid proposed MGF, two-sided exponential density, and transformations. Exercise 40 closely parallels HW5 Problem 4 with a different centering constant.
- **Covariance and dependence:** Exercises 42, 46–47, p.140. Use a table and test the false converse of independence implying zero covariance.
- **Linear combinations:** Exercises 49–55, p.140. These cover means, variances, covariance of two combinations, and sum/difference identities.

- **Conditional expectation:** Worked Example 20, p.139, then Exercise 60, p.140. Distinguish conditioning at a value from conditioning on an interval event.

14.2 Bertsekas and Tsitsiklis: intuition and derivations

The supplied file is *Introduction to Probability*, MIT 6.041–6.431 **Fall 2000 lecture notes**. Its page numbering restarts in each chapter; it is not the pagination of a later published edition. Use the section title or the following PDF-page ranges:

- **Expectation and variance:** Ch2 §2.4, chapter-local pp.11–22, PDF pp.66–77. Rework the examples with their solutions covered.
- **Discrete joint laws, conditioning, independence:** Ch2 §§2.5–2.7, chapter-local pp.22–42, PDF pp.77–97. Build and normalize conditional slices of tables.
- **Continuous joint laws:** Ch3 §3.5, chapter-local pp.26–39, PDF pp.125–138. Draw the support before reading each integral's limits.
- **MGFs:** Ch4 §4.1, chapter-local pp.2–13, PDF pp.153–164. Pay attention to the sign convention $\mathbb{E}[e^{sX}]$ and domain of the transform.
- **Conditional expectation as a random variable:** Ch4 §4.3, chapter-local pp.17–24, PDF pp.168–175. This supplies the tower rule and total variance as supplementary tools.
- **Covariance and correlation:** Ch4 §4.5, chapter-local pp.29–31, PDF pp.180–182. Read the geometric and linear-dependence meaning of correlation.

14.3 Casella and Berger: a more formal second pass

George Casella and Roger L. Berger, *Statistical Inference*, second edition, supplied scanned PDF. Use these verified section titles and their printed starting pages; the scan's text layer is not searchable, so navigate with page thumbnails if necessary.

- §2.2, *Expected Values*, p.55: formal properties of expectation.
- §2.3, *Moments and Moment Generating Functions*, p.59: moments, uniqueness, and why existence hypotheses matter.
- §4.1, *Joint and Marginal Distributions*, p.139, and §4.2, *Conditional Distributions and Independence*, p.147: a rigorous reinforcement of Topic A.
- §4.4, *Hierarchical Models and Mixture Distributions*, p.162: optional deeper conditional-expectation tools.
- §4.5, *Covariance and Correlation*, p.169: reinforce Topics E–F.

Read the definitions and one worked example in each selected section first; the more advanced statistical material elsewhere in the book is unnecessary for these homework methods.

14.4 Wasserman: a concise review

Larry Wasserman, *All of Statistics*, 2004, supplied PDF. Use §§2.5–2.8 for joint laws, marginals, independence, and conditional distributions; Chapter 3, §§3.1–3.3 for expectation, properties, variance and

covariance; §3.5 for conditional expectation; and §3.6 for MGFs. These sections start on printed pp.31, 33, 34, and 36, respectively; Chapter 3 begins on p.47, conditional expectation on p.54, and MGFs on p.56. This is most useful for a brief second explanation after Freund, not for replacing the homework calculations.

14.5 Feller: discrete reasoning and conceptual depth

William Feller, *An Introduction to Probability Theory and Its Applications*, Volume I, second edition (1957), supplied scan. Read ChV §1, *Conditional Probability*, p.114, and §3, *Stochastic Independence*, p.125. For this midterm's expectation topics, read ChIX §2, *Expectations*, p.220; §4, *The Variance*, p.227; §5, *Covariance; Variance of a Sum*, p.229; and §8, *The Correlation Coefficient*, p.236.

Feller ChXI begins on p.264 and studies generating functions for integer-valued variables. Its probability generating function $G(s) = \mathbb{E}(s^X)$ differs from an MGF: when both are defined, $M_X(t) = G(e^t)$. Read this as optional context; do not substitute $G'(0)$ for $M'(0)$ in your MGF work.

14.6 Course documents used for alignment

- *STA 431/631 Homework 4*, two-page supplied assignment: all common and STA 431 problems drive the joint-distribution models and region calculations.
- *STA 431/631 Homework 5*, two-page supplied assignment: all common and STA 431 problems drive the moment, MGF, covariance, and conditional-moment models.
- *STA431 Study Guide*, supplied 129-page whole-course PDF: Units 4–6, printed pp.32–61, supplied the initial reading structure. Its recommendations were checked against the owning textbooks where used; its calculations were not treated as an answer key for this guide.

The common and STA 431 problems are the main homework keys. Separately labeled STA 631 extensions are included for extra practice with a nonrectangular support and zero covariance without independence; they do not change the stated STA 431 scope.